# VF FOOTWEAR
# PERFORMANCE TESTING MANUAL

## Footwear
## *Standard Testing Performance*

**Effective from:** Season Spring 2026

©2026 VF Corporation. All Rights Reserved
Internal Use

# Table of Content

**TABLE OF CONTENT** ..... **2**

**SECTION 1: INTRODUCTION** ..... **4**

**SECTION 2: PERFORMANCE TESTING PROCEDURES** ..... **5**

2.1 VF Team's Responsibility ..... 5
2.2 Footwear Factory's Responsibility ..... 5
2.3 Third Party Testing / Nominated Laboratory's Responsibility ..... 7
2.4 Lab Test Procedures ..... 8
2.5 VF Contact List ..... 10

**SECTION 3: MATERIAL / PRODUCT SPECIFICATION MATRIX** ..... **11**

**SECTION 3: MANDATORY PERFORMANCE TEST PACKAGE FOR FOOTWEAR** ..... **12**

3.1 Fabric Upper Performance Package Test for Full Test and Additional Color (Specification 4.1) ..... 12
3.2 Fabric Lining Package Test for Full Test and Additional Color (Specification 4.2) ..... 13
3.3 Leather Upper Performance Package Test for Full Test (Specification 4.3) ..... 14
3.4 Leather Lining Performance Package Test for Full Test (Specification 4.4) ..... 15
3.5 Synthetic Leather Upper Performance Package Test for Full Test and Additional Color (Specification 4.5) ..... 16
3.6 Synthetic Leather Lining Performance Package Test for Full Test and Additional Color (Specification 4.6) ..... 17
3.7 Hardware Performance Package Test for Full Test (Specification 4.7) ..... 17
3.8 Rubber Performance Package Test for Full Test (Specification 4.8, 4.9,4.10) ..... 18
3.9 EVA Performance Package Test for Full Test (Specification 4.11, 4.12) ..... 18
3.10 TPU Package Test for Full Test Performance (Specification 4.13) ..... 19
3.11 PU Footbed Package Test for Full Test Performance (Specification 4.14) ..... 19
3.12 Sockliner Performance Package Test for Full Test (Specification 4.15) ..... 20
3.13 PU Foam Performance Package Test for Full Test (Specification 4.16) ..... 20
3.14 Webbing Performance Package Test for Full Test and Additional Color (Specification 4.17) ..... 21
3.15 Lace Performance Package Test for Full Test and Additional Color (Specification 4.18) ..... 21
3.16 Hook & Loop (Velcro®) Performance Package Test for Full Test and Additional Color (Specification 4.19) ..... 22
3.17 Thread Performance Package Test for Full Test and Additional Color (Specification 4.20) ..... 22
3.18 Upper Trim Performance Package Test for Full Test (Specification 4.21) ..... 23
3.19 Zipper Performance Package Test for Full Test and Additional Color (Specification 4.22) ..... 23
3.20 Straps & Plugs Performance Package Test for Full Test (Specification 4.23) ..... 24
3.21 Ink & Printed Material Performance Package Test for Full Test (Specification 4.24) ..... 24
3.22 Insole Performance Package Test for Full Test (Specification 4.25) ..... 25
3.23 Elastic Cord Performance Package Test for Full Test and Additional Color (Specification 4.26) ..... 25
3.24 Wrapping Paper Performance Package Test for Full Test (Specification 4.27) ..... 26
3.25 Shoe Box Performance Package Test for Full Test (Specification 4.28) ..... 26
3.26 Carton Performance Package Test for Full Test (Specification 4.28) ..... 26
3.27 Woven Label Performance Package Test for Full Test and Additional Color (Specification 4.29) ..... 27
3.28 Reinforcement Performance Package Test for Full Test and Additional Color (Specification 4.30) ..... 27
3.29 Shoe Charms Performance Package Test for Full Test (Specification 4.31) ..... 28
3.30 Whole Shoe Performance Package Test for Full Test (Specification 4.32, 4.33, 4.34 & 4.35&4.36) ..... 28

©2026 VF Corporation. All Rights Reserved
Internal Use

**SECTION 4: TESTING REQUIREMENT ..... 29**

4.1 Fabric Upper Test Requirement ..... 29
4.2 Fabric Lining Test Requirement ..... 32
4.3 Leather Upper Test Requirement ..... 34
4.4 Leather Lining Test Requirement ..... 37
4.5 Synthetic Leather Upper Test Requirement ..... 39
4.6 Synthetic Leather Lining Test Requirement ..... 42
4.7 Hardware Test Requirement ..... 45
4.8 Rubber Test Requirement ..... 46
4.9 Blown Rubber Test Requirement ..... 48
4.10 Foxing Tape Test Requirement ..... 50
4.11 EVA Midsole Test Requirement ..... 51
4.12 EVA Outsole Test Requirement ..... 53
4.13 TPU Outsole Test Requirement ..... 54
4.14 PU Footbed Test Requirement ..... 55
4.15 Sock liner (EVA – Latex – PU) Test Requirement ..... 57
4.16 PU Foam Test Requirement ..... 59
4.17 Webbing Test Requirement ..... 60
4.18 Lace (Cotton – Synthetic) Test Requirement ..... 61
4.19 Hook & Loop (Velcro ®) Test Requirement ..... 63
4.20 Thread Test Requirement ..... 64
4.21 Upper Trim Test Requirement ..... 65
4.22 Zipper Test Requirement ..... 66
4.23 Straps & Plugs Test Requirement ..... 68
4.24 Ink & Print Material Test Requirement ..... 70
4.25 Insole Test Requirement ..... 72
4.26 Elastic Cord Test Requirement ..... 73
4.27 Wrapping Paper Test Requirement ..... 74
4.28 Carton – Shoe Box Test Requirement ..... 75
4.29 Woven Label Test Requirement ..... 76
4.30 Reinforcement Test Requirement ..... 77
4.31 Shoe Charms Test Requirement ..... 79
4.32 Bonding Strength Test Requirement ..... 80
4.33 Bonding Strength Tolerance ..... 82
4.34 Pull Strength Test Requirement ..... 84
4.35 Whole Shoe Performance Test Requirement ..... 85
4.36 Snow Boots (Vans) Performance Test Requirement ..... 87

**LEATHER TANNAGE SPECIFICATION ..... 88**

**ONE VF POLYMER MATRIX ..... 88**

**SECTION 5: SAMPLING FOR TESTING ..... 89**

5.1 Mass Production Testing Frequency (Factory In-house lab) ..... 89

**SECTION 6: NOMINATED TESTING LABORATORIES ..... 90**

**SECTION 7: CHANGE LOG ..... 92**

©2026 VF Corporation. All Rights Reserved
Internal Use

# Section 1: Introduction

This Performance Testing Manual is designed to ensure that a uniform understanding and implementation of VF Corporation Quality Assurance program by all offices and laboratories. All suppliers are required to submit their bulk samples to nominated third party testing laboratories for testing. The list of nominated third party testing laboratories is provided in Section 6 of this manual. VFC Lab will provide the service mainly on development stage and cross checking. The manual covers the responsibility of all concerned parties and guidelines of the Quality testing procedure for VF suppliers in conjunction with VF RSL Implementation Manual. These are the minimum requirements that must be met before using for VF Brands' production.

In addition to VF Brand Quality Testing Manual, suppliers are also responsible for following ***VF RSL Implementation Manual*** to perform the required Restricted Substances List (RSL) test for VF to validate product compliance. However, materials and products which only meet the RSL tests listed in the VF RSL Implementation manual may not equivalent to VF RSL compliance.

Each supplier of VF Product or Raw Material represents and warrants that each of its product or raw materials (whether a VF Product or Raw Material) complies with all provisions of the RSL (including, but not limited to, the RSL prohibitions, restrictions and other requirements) and that the supplier agrees to indemnify and hold harmless VF Corporation and its subsidiaries and brands (collectively, "VF") from any claim, loss, damage or other detriment, resulting from any such supplier's non-compliance.

**Note: This document does not cover the regulatory requirement for emerging markets, including but not limit to Egypt, Saudi Arabia, Morocco, which would have specific testing and quality inspection requirements on products and product manufacturers. Please seek advice from the VF Global Product Stewardship team. Do not rely exclusively on this guideline as the only source for determining product standards.**

Prepared by:

Global Product Stewardship Team

©2026 VF Corporation. All Rights Reserved
Internal Use

# Section 2: Performance Testing Procedures

## 2.1 VF Team's Responsibility

1. Provide to material supplier and footwear factory a copy of VF Footwear Performance Testing Manual, VF RSL and VF RSL Implementation Manual, sample submission requirements and performance standards to suppliers.
2. Select the accredited / nominated testing laboratories.
3. Provide technical comments on the lab test report

## 2.2 Footwear Factory's Responsibility

1. Prepare test samples from the bulk production materials.
2. Follow the lab test procedure to create online test request on SARGE (https://asiaportal.ecvision.com/vf)[1], print out *fully completed* test request form and send along with the samples to the accredited or nominated laboratory.
3. Material can only be used upon the testing result is approved by the VF through SARGE.
4. Factory has the responsibility to present all the testing results to VF team.
5. Testing result is invalid when the submitted test sample is not representing the bulk quality from production.
6. For any failed test report, fabric supplier is responsible to remediate the fabric and re-test following the same requirement and procedure. Re-testing must be performed at the same laboratory as the original test submission.
7. Footwear factory / material suppliers are responsible for all testing cost.
8. Finished goods makers have full responsibility on the quality compliance of factory-sourced fabrics and should follow up the testing under the same lab test procedure.

**Note: If a footwear factory does not have an accredited in-house lab under VF factory in-house lab accreditation program, each style of footwear or material must be tested in a nominated 3rd party lab at least every 10,000 units or pairs.**

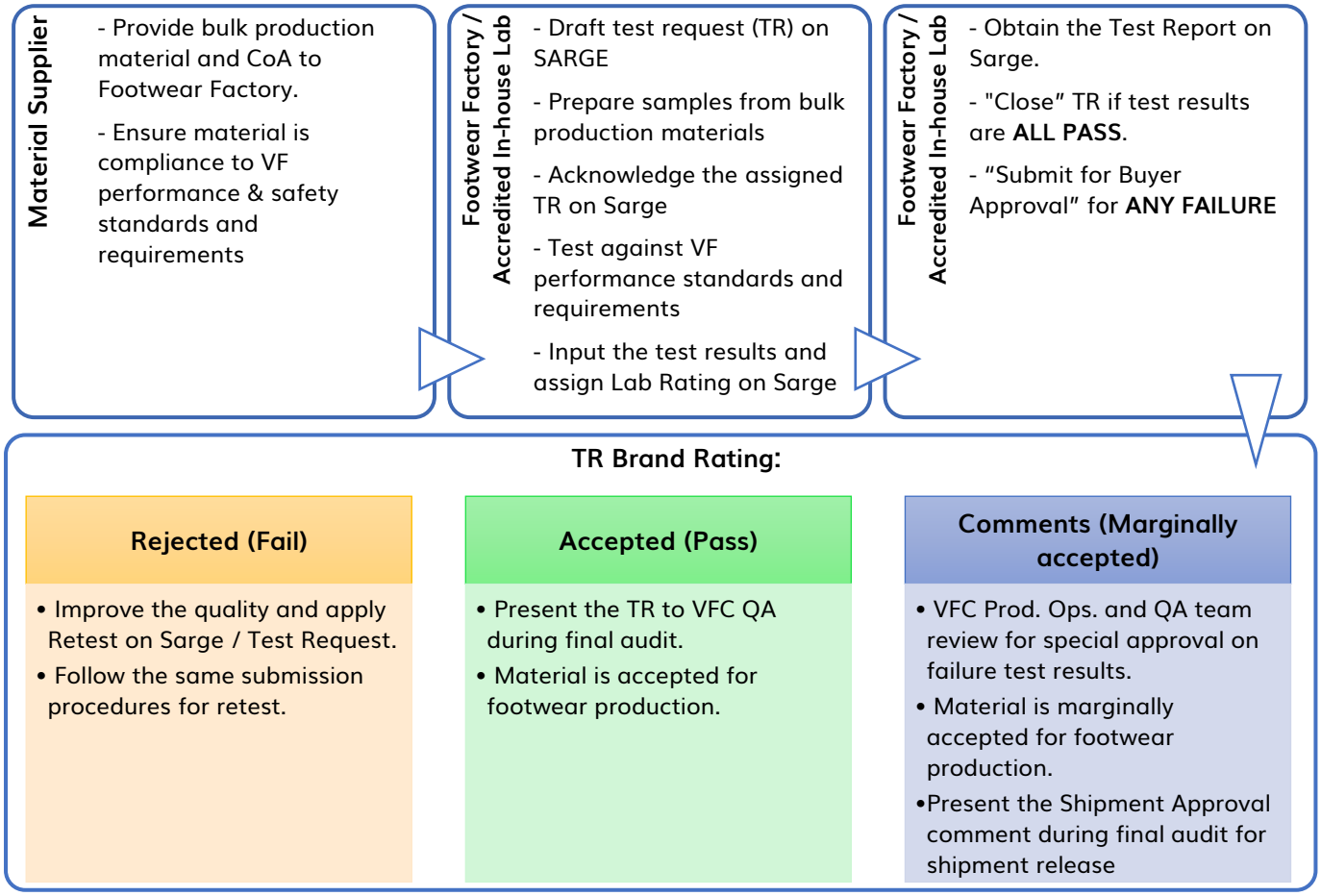
**No shipment is allowed without VF Quality Team approval**

[1] For EMEA: https://portal.ecvision.com/vf

©2026 VF Corporation. All Rights Reserved
Internal Use

# Flow Chart for Footwear Factory:

*All test application and shipment approval processes are done online on SARGE at https://asiaportal.ecvision.com/vf*
*Please communicate with VF Sourcing for SARGE user guide.*




**Sampling test for material / finished footwear:**

- For new material, one colorway is necessary for full package testing; and other colorways are necessary for the additional color testing (See Section 3).
- Most of Restricted Substance List (RSL) tests are valid for 12 calendar months, while Chromium (VI) Content test for leather is valid for 6 calendar months only. For factory-sourced material report is valid for one season only.
- VF reserves the right to change the testing frequency.

©2026 VF Corporation. All Rights Reserved
Internal Use

## 2.3 Third Party Testing / Nominated Laboratory's Responsibility

1. Pick up test samples from vendor / supplier.
2. Contact the applicants when information on the online test request form is incomplete / uncertain.
3. Contact the applicants when the submitted samples are visually unsatisfactory (e.g. stained, torn, missing pieces, etc.).
4. Perform all testing items listed in the packages, select the correct testing specification standard as testing requirement and indicate the testing package / specification number on test reports.
5. Add sample photos in original state on test report.
6. In case of any failure on "Appearance" or "Restricted Substance List" test, add photo of which component(s) with failure with appropriate indication on test reports.
7. Input the test results and lab ratings on SARGE.
8. Distribute the test reports to applicants and appointed notified party. Email the test report in PDF format and invoice to applicant.
9. Store the original testing samples and testing specimens for 6 months after completion of testing.

©2026 VF Corporation. All Rights Reserved
Internal Use

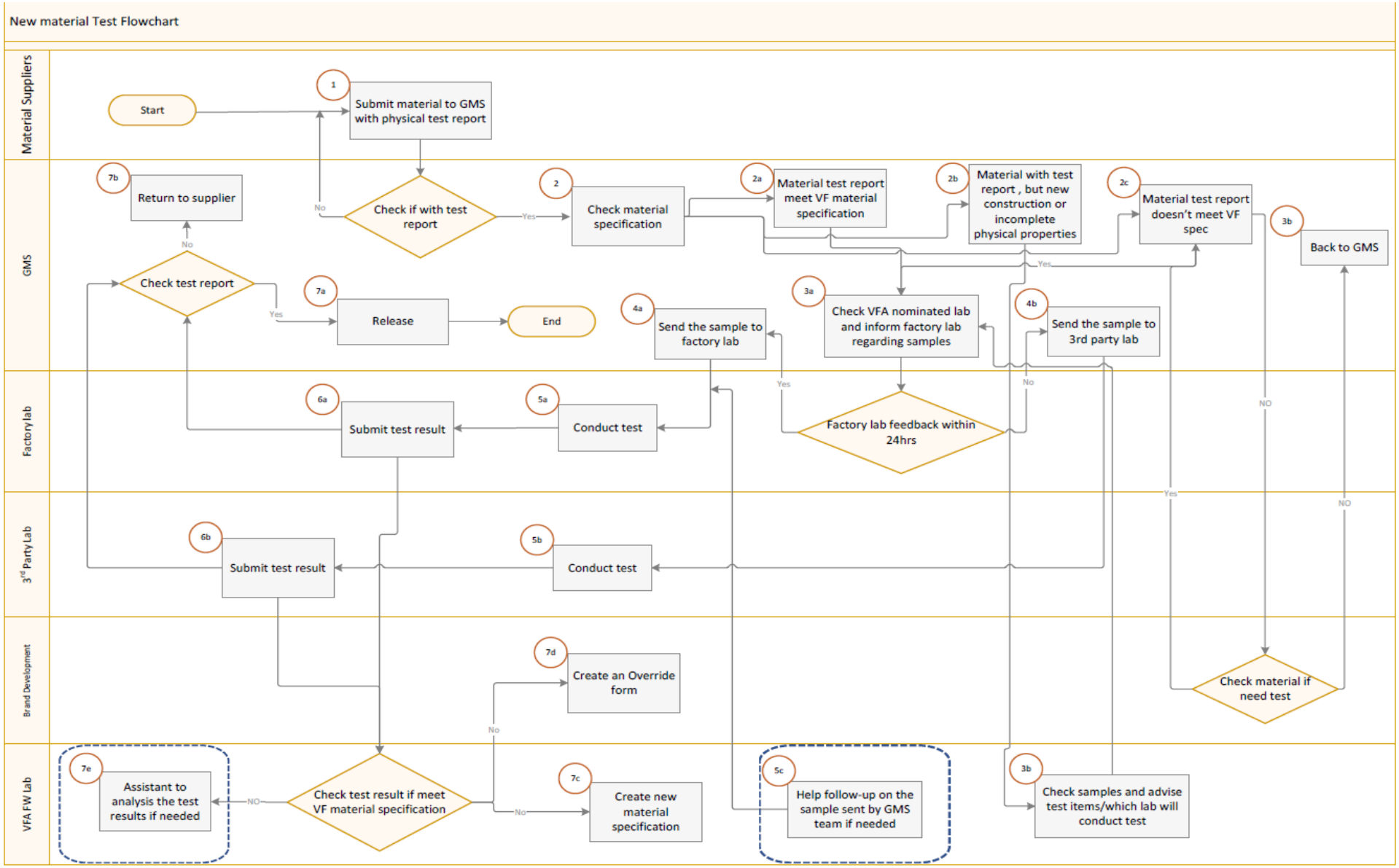
## 2.4 Lab Test Procedures

Please complete the Test Request Form by selecting or inputting all required information as follows:

- Testing Lab Information
- Applicant / Supplier Information
- Notify Email Group *(i.e. Email address of related merchandiser)*
- Sample Description *(i.e. Specification of submitted test sample, e.g. season, stage, end use, weight, finishing, requirement specification no., estimated test sample submission date, etc.)*
- Brand Information
- Product / Material Information *(i.e. Style / material no., color code, color name, submitted test sample quantity, etc.)*
  *Important Note: Input Style No. for finished goods test, or Material / Trim No. for raw material test.*
- Previous Test Report Number *(e.g. Full Test Report Number for additional color test submission)*

Previous RSL Test Report Number *(e.g. Fabric Flammability Test Report Number, other CPSIA corresponding test reports)*

©2026 VF Corporation. All Rights Reserved
Internal Use




©2026 VF Corporation. All Rights Reserved
Internal Use

## 2.5 VF Contact List

### VF Asia Sourcing Limited

**VFA Zhongshan Office – Footwear:**
**Cathy Zhou** email: Cathy_Zhou@vfc.com
1F, Block 5, No.2, 1st Shiji Road, HuoJu Development Zone, Zhongshan City, GuangDong, PRC 528437
Tel: (86) 0760 8734 1482

**VFA Taiwan Office:**
**Bow Lin** e-mail: Bow_Lin@vfc.com
Tel: +(886)2 8178 7053

### VF Latin America Sourcing S.A.R.L – Honduras

**Olga Ayala** e-mail: Olga_Ayala@vfc.com
**Zoli** Honduras, KM 5, Carretera a Puerto Cortes Choloma, Cortes, Honduras, C.A.
Tel: + 011-504-2606-3192 Ext. 230

### VF EUROPE

**Pierre Ierardi** e-mail: pierre_ierardi@vfc.com
Tel: +41.91.649.1519

©2026 VF Corporation. All Rights Reserved
Internal Use

# Section 3: Material / Product Specification Matrix

| | Material | | | |
|---|---|---|---|---|
| | Fabric / Textile | Leather | Synthetic Leather | Others |
| Upper | 3.1 & 4.1 | 3.3 & 4.3 | 3.5 & 4.5 | |
| Lining | 3.2 & 4.2 | 3.4 & 4.4 | 3.6 & 4.6 | |
| Hardware | | | | 3.7 & 4.7 |
| Webbing | | | | 3.14 & 4.17 |
| Lace | | | | 3.15 & 4.18 |
| Velcro | | | | 3.16 & 4.19 |
| Thread | | | | 3.17 & 4.20 |
| Insole | | | | 3.22 & 4.25 |
| Ink-Print material | | | | 3.21 & 4.24 |
| Sock liner | | | | 3.12 & 4.15 |
| PU Foam | | | | 3.13 & 4.16 |
| Zipper | | | | 3.19 & 4.22 |
| Strap and Plugs | | | | 3.20& 4.23 |
| Upper Trim | | | | 3.18 & 4.21 |
| Elastic cord | | | | 3.23 & 4.26 |
| Wrapping Paper | | | | 3.24 & 4.27 |
| Shoe Box | | | | 3.25 & 4.28 |
| Carton | | | | 3.26 & 4.28 |
| Woven label | | | | 3.27 & 4.29 |
| Shoe Charms | | | | 3.29 & 4.31 |
| Reinforcement | | | | 3.28 & 4.30 |

| | Material | | | |
|---|---|---|---|---|
| | Rubber | EVA | TPU | PU |
| Midsole | | 3.9 & 4.11 | | |
| Outsole | 3.8 & 4.8 4.9 | 3.9 & 4.12 | 3.10 & 4.13 | |
| Foxing Tape | 3.8 & 4.10 | | | |
| Footbed | | | | 3.11 & 4.14 |

| Finished Shoe | |
|---|---|
| Bonding | 3.30 & 4.32, 4.33 |
| Pull Strength | 3.30 & 4.34 |
| Whole shoe Performance | 3.30 & 4.35 |
| Snow Boot | 3.30 & 4.36 |

Remark
Section 3 – Mandatory Performance Test Package for Footwear
Section 4 – Testing Requirement

©2026 VF Corporation. All Rights Reserved
Internal Use

# Section 3: Mandatory Performance Test Package for Footwear

**X** – Mandatory test under this test package; **(X)** – test is required if applicable

## 3.1 Fabric Upper Performance Package Test for Full Test and Additional Color (Specification 4.1)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 02 | Elongation | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 03 | Trouser Tear | X | | VF-B07 | ASTM D2261-13(2024) | ✓ |
| 04 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 05 | Bursting | X | | VF-B02 | ASTM D3786/D3786M-18(2023) | ✓ |
| 06 | Martindale Abrasion | (X) | | VF-C04 | N/A | ✓ |
| 07 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 08 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |
| 09 | Color Fastness to Perspiration (for direct skin contact only) | (X) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 10 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 11 | Heat Aging | (X) | | VF-E04 | N/A | ✓ |
| 12 | Hydrolysis (For printed and coated material) | (X) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 13 | Maeser Test (water resistant material) | (X) | (X) | VF-E11 | N/A | ✓ |
| 14 | Dynamic Absorption (water resistant material) | (X) | | VF-E11 | N/A | ✓ |
| 15 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 16 | Static Water Absorption (water resistant material) | (X) | | VF-E10 | N/A | ✓ |
| 17 | Yarn Count | (X) | | VF-A07 | ASTM D3775-17(2023) | ✓ |
| 18 | Weight | (X) | | VF-A02 | ASTM D3776/D3776M-20(2025) | ✓ |
| 19 | Thickness | X | | VF-A03 | N/A | ✓ |
| 20 | Peel Strength to Rubber | (X) | | VF-F02 | N/A | ✓ |
| 21 | Laminar Peel Strength | (X) | | VF-F04 | N/A | ✓ |
| 22 | Finish Adhesion | (X) | | VF-F03 | N/A | ✓ |
| 23 | pH Value | X | | ISO 3071 | ISO 3071 | ✓ |
| 24 | Flammability test | X | | CAN/CGSB 4.2 No. 27.5 | CAN/CGSB 4.2 No. 27.5 | ✓ |
| 25 | Spray Test | (X) | | VF-E27 | ISO 4920 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.2 Fabric Lining Package Test for Full Test and Additional Color (Specification 4.2)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 02 | Elongation | X | | VF-B01 | ASTM D5035-11(2024 | ✓ |
| 03 | Trouser Tear | X | | VF-B07 | ASTM D2261-13(2024) | ✓ |
| 04 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 05 | Bursting | X | | VF-B02 | ASTM D3786/D3786M-18(2023) | ✓ |
| 06 | Martindale Abrasion | X | | VF-C04 | N/A | ✓ |
| 07 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 08 | Color Fastness to Perspiration | X | X | VF-E16 | AATCC TM15-2021 | ✓ |
| 09 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 10 | Heat Aging | (X) | | VF-E04 | N/A | ✓ |
| 11 | Hydrolysis (For printed and coated material) | (X) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 12 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 13 | Static Water Absorption | (X) | | VF-E10 | N/A | ✓ |
| 14 | Yarn Count | (X) | | VF-A07 | ASTM D3775-17(2023) | ✓ |
| 15 | Weight | (X) | | VF-A02 | ASTM D3776/D3776M-20(2025) | ✓ |
| 16 | Thickness | X | | VF-A03 | N/A | ✓ |
| 17 | pH Value | X | | ISO 3071 | ISO 3071 | ✓ |
| 18 | Color Fastness to Water | (X) | (X) | VF-E02 | N/A | ✓ |
| 19 | Flammability test | X | | CAN/CGSB 4.2 No. 27.5 | CAN/CGSB 4.2 No. 27.5 | ✓ |
| 20 | Spray Test | (X) | | VF-E27 | ISO 4920 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.3 Leather Upper Performance Package Test for Full Test (Specification 4.3)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile Strength | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Stitch Tear | (X) | VF-B08 | SATRA TM33 | ✓ |
| 04 | Baumann Tear | X | VF-B09 | ISO 3377-2 | ✓ |
| 05 | Bursting | X | VF-B02 | N/A | ✓ |
| 06 | Bally flex | (X) | VF-D01 | ASTM D6182-23 | ✓ |
| 07 | Taber Abrasion | (X) | VF-C05 | ASTM D7255-22e1(2022) | ✓ |
| 08 | Color Fastness to Crocking | X | VF-E02 | ISO 20433:2024(E) | ✓ |
| 09 | Color Fastness to Water | X | VF-E01 | N/A | ✓ |
| 10 | Color Fastness to Perspiration (for direct skin contact only) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 11 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 12 | Heat Aging | (X) | VF-E04 | N/A | ✓ |
| 13 | Hydrolysis (For PU coated leather) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 14 | Maeser Test (water resistant material) | (X) | VF-E11 | N/A | ✓ |
| 15 | Dynamic Absorption (water resistant material) | (X) | VF-E11 | N/A | ✓ |
| 16 | Water wicking | (X) | VF-E14 | N/A | ✓ |
| 17 | Static Water Absorption (water resistant material) | (X) | VF-E10 | ASTM D6015-21(2022) | ✓ |
| 18 | Oil Content | (X) | VF-I01 | N/A | ✓ |
| 19 | Weight | (X) | VF-A02 | N/A | ✓ |
| 20 | Thickness | X | VF-A03 | ASTM D1813-13(2023) | ✓ |
| 21 | Peel Strength to Rubber | (X) | VF-F02 | N/A | ✓ |
| 22 | Laminar Peel Strength | (X) | VF-F04 | N/A | ✓ |
| 23 | Finish Adhesion | (X) | VF-F03 | N/A | ✓ |
| 24 | Tape Test | (X) | VF-F05 | N/A | ✓ |
| 25 | PVC Migration | (X) | VF-E06 | ISO 15701-2022 | ✓ |
| 26 | pH Value | X | ISO 4045 | ISO 4045 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.4 Leather Lining Performance Package Test for Full Test (Specification 4.4)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile Strength | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Stitch Tear | (X) | VF-B08 | SATRA TM33 | ✓ |
| 04 | Baumann Tear | X | VF-B09 | ISO 3377-2 | ✓ |
| 05 | Bursting | X | VF-B02 | N/A | ✓ |
| 06 | Bally flex | (X) | VF-D01 | ASTM D6182-23 | ✓ |
| 07 | Martindale Abrasion | X | VF-C04 | N/A | ✓ |
| 08 | Color Fastness to Crocking | X | VF-E02 | ISO 20433: 2024(E) | ✓ |
| 09 | Color Fastness to Water | (X) | VF-E01 | N/A | |
| 10 | Color Fastness to Perspiration | X | VF-E16 | AATCC TM15-2021 | ✓ |
| 11 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 12 | Heat Aging | (X) | VF-E04 | N/A | ✓ |
| 13 | Hydrolysis (For PU coated leather and print leather) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 14 | Water wicking | (X) | VF-E14 | N/A | ✓ |
| 15 | Static Water Absorption | (X) | VF-E10 | ASTM D6015-21(2022) | ✓ |
| 16 | Finish Adhesion | (X) | VF-F03 | N/A | ✓ |
| 17 | Tape Test | (X) | VF-F05 | N/A | ✓ |
| 18 | Oil Content | (X) | VF-I01 | N/A | ✓ |
| 19 | Weight | (X) | VF-A02 | N/A | ✓ |
| 20 | Thickness | X | VF-A03 | ASTM D 1813-13(2023) | ✓ |
| 21 | PVC Migration | (X) | VF-E06 | ISO 15701-2022 | ✓ |
| 22 | pH Value | X | ISO 4045 | ISO 4045 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.5 Synthetic Leather Upper Performance Package Test for Full Test and Additional Color (Specification 4.5)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 02 | Elongation | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 03 | Trouser Tear | X | | VF-B07 | ASTM D2261-13(2024) | ✓ |
| 04 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 05 | Bursting | X | | VF-B02 | ASTM D3786/D3786M-18(2023) | ✓ |
| 06 | Bally flex | (X) | | VF-D01 | ASTM D6182-23 | ✓ |
| 07 | Taber Abrasion | (X) | | VF-C05 | ASTM D7255-22e1(2022) | ✓ |
| 08 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |
| 09 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 10 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 11 | Heat Aging | (X) | | VF-E04 | N/A | ✓ |
| 12 | Hydrolysis (For PU coated synthetic leather) | (X) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 13 | Maeser | (X) | (X) | VF-E11 | N/A | ✓ |
| 14 | Dynamic Absorption | (X) | | VF-E11 | N/A | ✓ |
| 15 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 16 | Static Water Absorption | (X) | | VF-E10 | N/A | ✓ |
| 17 | Weight | (X) | | VF-A02 | ASTM D3776/D3776M-20(2025) | ✓ |
| 18 | Thickness | X | | VF-A03 | N/A | ✓ |
| 19 | Peel Strength to Rubber | (X) | | VF-F02 | N/A | ✓ |
| 20 | Laminar Peel Strength | (X) | | VF-F04 | N/A | ✓ |
| 21 | Finish Adhesion | (X) | | VF-F03 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.6 Synthetic Leather Lining Performance Package Test for Full Test and Additional Color (Specification 4.6)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 02 | Elongation | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 03 | Trouser Tear | X | | VF-B07 | ASTM D2261-13(2024) | ✓ |
| 04 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 05 | Bursting | X | | VF-B02 | ASTM D3786/D3786M-18(2023) | ✓ |
| 06 | Bally flex | (X) | | VF-D01 | ASTM D6182-23 | ✓ |
| 07 | Martindale Abrasion | X | | VF-C04 | N/A | ✓ |
| 08 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 09 | Color Fastness to Water | (X) | | VF-E01 | N/A | |
| 10 | Color Fastness to Perspiration | X | X | VF-E16 | AATCC TM15-2021 | ✓ |
| 11 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 12 | Heat Aging | (X) | | VF-E04 | N/A | ✓ |
| 13 | Hydrolysis (For PU coated synthetic leather) | (X) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 14 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 15 | Static Water Absorption | (X) | | VF-E10 | N/A | ✓ |
| 16 | Finish Adhesion | (X) | | VF-F03 | N/A | ✓ |
| 17 | Weight | (X) | | VF-A02 | ASTM D3776/D3776M-20(2025) | ✓ |
| 18 | Thickness | X | | VF-A03 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

## 3.7 Hardware Performance Package Test for Full Test (Specification 4.7)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Corrosion Test | X | VF-E09 | N/A | ✓ |
| 02 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 03 | Hydrolysis | (X) | VF-E19 | SATRA TM344 | ✓ |
| 04 | Heat aging | X | VF-E04 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.8 Rubber Performance Package Test for Full Test (Specification 4.8, 4.9,4.10)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile Strength | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Die C Tear | X | VF-B05 | ASTM D624-00(2020) | ✓ |
| 04 | Compression Set | (X) | VF-B10 | N/A | ✓ |
| 05 | Seam Stitch Tear-Soiling | (X) | VF-B20 | N/A | ✓ |
| 06 | Ross Flex | X | VF-D02 | N/A | ✓ |
| 07 | DIN Abrasion (for ground contact only) | (X) | VF-C01 | N/A | ✓ |
| 08 | Shrinkage | X | VF-E13 | N/A | ✓ |
| 09 | Blooming | X | VF-E17 | SATRA TM361 | ✓ |
| 10 | Hydrolysis | X | VF-E19 | SATRA TM344 | ✓ |
| 11 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 12 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 13 | Hardness | X | VF-A01 | N/A | ✓ |
| 14 | Specific Gravity | X | VF-A04 | ASTM D792-20 | ✓ |
| 15 | Coefficient of Friction -TM144 (for ground contact only) | (X) | VF-B25 | SATRA TM144 | ✓ |
| 16 | Non- Marking | (X) | VF-E21 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.9 EVA Performance Package Test for Full Test (Specification 4.11, 4.12)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Die C Tear | X | VF-B05 | N/A | ✓ |
| 04 | Split Tear | X | VF-B06 | N/A | ✓ |
| 05 | Seam Stitch Tear-Soiling | (X) | VF-B20 | N/A | ✓ |
| 06 | Static Compression Set | X | VF-B10 | N/A | ✓ |
| 07 | Resilience | X | VF-B16 | ISO 4662:2017 | ✓ |
| 08 | Ross Flex | X | VF-D02 | N/A | ✓ |
| 09 | DIN Abrasion (for ground contact only) | (X) | VF-C01 | N/A | ✓ |
| 10 | Shrinkage | X | VF-E13 | N/A | ✓ |
| 11 | Blooming | (X) | VF-E17 | SATRA TM361 | ✓ |
| 12 | Color Fastness to Crocking | (X) | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 13 | Color Fastness to Perspiration (for direct skin contact only) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 14 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 15 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 16 | Hydrolysis | (X) | VF-E19 | SATRA TM344 | ✓ |

©2026 VF Corporation. All Rights Reserved

Internal Use

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 17 | Hardness | X | VF-A01 | N/A | ✓ |
| 18 | Specific Gravity | X | VF-A04 | ASTM D792-20 | ✓ |
| 19 | Coefficient of Friction -TM144 (for ground contact only) | (X) | VF-B25 | SATRA TM144 | ✓ |
| 20 | Peel Strength | (X) | VF-F02 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.10 TPU Package Test for Full Test Performance (Specification 4.13)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile Strength | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Die C Tear | X | VF-B05 | ASTM D624-00(2020) | ✓ |
| 04 | Ross Flex | X | VF-D02 | N/A | ✓ |
| 05 | DIN Abrasion | (X) | VF-C01 | N/A | ✓ |
| 06 | Shrinkage | X | VF-E13 | N/A | ✓ |
| 07 | Blooming | (X) | VF-E17 | SATRA TM361 | ✓ |
| 08 | Hydrolysis | (X) | VF-E19 | SATRA TM344 | ✓ |
| 09 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 10 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 11 | Hardness | X | VF-A01 | N/A | ✓ |
| 12 | Specific Gravity | X | VF-A04 | ASTM D792-20 | ✓ |
| 13 | Coefficient of Friction -TM144 (for ground contact only) | (X) | VF-B25 | SATRA TM144 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.11 PU Footbed Package Test for Full Test Performance (Specification 4.14)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile Strength | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Die C Tear | X | VF-B05 | N/A | ✓ |
| 04 | Split Tear | X | VF-B06 | N/A | ✓ |
| 05 | Static Compression Set | X | VF-B10 | N/A | ✓ |
| 06 | Resilience | X | VF-B16 | ISO 4662:2017 | ✓ |
| 07 | Ross Flex | X | VF-D02 | N/A | ✓ |
| 08 | Shrinkage | X | VF-E13 | N/A | ✓ |
| 09 | Blooming | (X) | VF-E17 | SATRA TM361 | ✓ |
| 10 | Hydrolysis | (X) | VF-E19 | SATRA TM344 | ✓ |
| 11 | Color Fastness to Crocking | (X) | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 12 | Color Fastness to Perspiration (for direct skin contact only) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 13 | Color Fastness to Light | (X) | VF-E03 | N/A | ✓ |
| 14 | Heat aging | (X) | VF-E04 | N/A | ✓ |

©2026 VF Corporation. All Rights Reserved
Internal Use

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 15 | Non marking | (X) | VF-E21 | N/A | ✓ |
| 16 | Hardness | X | VF-A01 | N/A | ✓ |
| 17 | Specific Gravity | X | VF-A04 | ASTM D792-20 | ✓ |
| 18 | Peel Strength | (X) | VF-F02 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.12 Sockliner Performance Package Test for Full Test (Specification 4.15)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile Strength | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Die C Tear | X | VF-B05 | N/A | ✓ |
| 04 | Split Tear | X | VF-B06 | N/A | ✓ |
| 05 | Static Compression Set | X | VF-B10 | N/A | ✓ |
| 06 | Resilience | X | VF-B16 | ISO 4662:2017 | ✓ |
| 07 | Shrinkage | X | VF-E13 | N/A | ✓ |
| 08 | Color Fastness to Crocking | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 09 | Color Fastness to Perspiration | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 10 | Color Fastness to Light (for exposure part) | (X) | VF-E03 | N/A | ✓ |
| 11 | Hardness | X | VF-A01 | N/A | ✓ |
| 12 | Specific Gravity | X | VF-A04 | ASTM D792-20; ASTM D3574-17 | ✓ |
| 13 | Laminar Peel Strength | X | VF-F04 | N/A | ✓ |
| 14 | Ink Adhesion on Substrate | (X) | VF-F05 | N/A | ✓ |
| 15 | Martindale Abrasion | (X) | VF-C04 | N/A | ✓ |
| 16 | Hydrolysis | (X) | VF-E19 | SATRA TM344 | ✓ |
| 17 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 18 | pH Value (for natural leather and textile sockliner cover) | X | ISO 3071, ISO 4045 | ISO 3071, ISO 4045 | ✓ |
| 19 | Flammability test | X | CAN/CGSB 4.2 No. 27.5 | CAN/CGSB 4.2 No. 27.5 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.13 PU Foam Performance Package Test for Full Test (Specification 4.16)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile Strength | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Die-C Tear | X | VF-B05 | N/A | ✓ |
| 04 | Split Tear | X | VF-B06 | N/A | ✓ |
| 05 | Static Compression Set | X | VF-E10 | N/A | ✓ |
| 06 | Color Fastness to Light (exposure part only) | (X) | VF-E03 | N/A | ✓ |
| 07 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 08 | Hydrolysis | (X) | VF-E19 | SATRA TM344 | ✓ |
| 09 | Hardness | X | VF-A01 | N/A | ✓ |

©2026 VF Corporation. All Rights Reserved

Internal Use

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 10 | Specific Gravity | X | VF-A04 | ASTM D3574 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.14 Webbing Performance Package Test for Full Test and Additional Color (Specification 4.17)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D6775-13(2024) | ✓ |
| 02 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 03 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 04 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |
| 05 | Color Fastness to Perspiration (for direct skin contact only) | (X) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 06 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 07 | Heat aging | (X) | (X) | VF-E04 | N/A | ✓ |
| 08 | Hydrolysis (For printed/coated material) | (X) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 09 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 10 | pH Value | X | | ISO 3071 | ISO 3071 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

## 3.15 Lace Performance Package Test for Full Test and Additional Color (Specification 4.18)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength (Wet test use for leather lace) | X | | VF-B01 | SATRA TM94 | ✓ |
| 02 | Tensile after abrasion | X | | VF-B01 | SATRA TM94 N/A | ✓ |
| 03 | Knot Slip Resistance | (X) | | VF-B23 | SATRA TM195 | ✓ |
| 04 | Lace Tip Retention | (X) | | VF-B24 | SATRA TM175 | ✓ |
| 05 | Lace Abrasion | X | | VF-C09 | N/A | ✓ |
| 06 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 07 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |
| 08 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 09 | Heat aging | (X) | (X) | VF-E04 | N/A | ✓ |
| 10 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 11 | pH Value | X | | ISO 3071, ISO 4045 | ISO 3071, ISO 4045 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.16 Hook & Loop (Velcro®) Performance Package Test for Full Test and Additional Color (Specification 4.19)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 02 | Elongation | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 03 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 04 | Peel Strength | X | | VF-B03 | SATRA TM123 | ✓ |
| 05 | Shear Strength | X | | VF-B03 | SATRA TM123 | ✓ |
| 06 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 07 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |
| 08 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 09 | Heat aging | (X) | (X) | VF-E04 | N/A | ✓ |
| 10 | pH Value | X | | ISO 3071 | ISO 3071 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

## 3.17 Thread Performance Package Test for Full Test and Additional Color (Specification 4.20)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | | VF-B01 | N/A | ✓ |
| 03 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 04 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |
| 05 | Color Fastness to Perspiration (for direct skin contact only) | (X) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 06 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 07 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 08 | Heat aging | (X) | (X) | VF-E04 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.18 Upper Trim Performance Package Test for Full Test (Specification 4.21)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile | X | VF-B01 | N/A | ✓ |
| 02 | Elongation | X | VF-B01 | N/A | ✓ |
| 03 | Die C Tear | X | VF-B05 | ASTM D624-00(2020) | ✓ |
| 04 | Stitch Tear | (X) | VF-B08 | SATRA TM33 | ✓ |
| 05 | Bally Flex | (X) | VF-D01 | ASTM D6182-23 | ✓ |
| 06 | Martindale Abrasion | (X) | VF-C04 | N/A | ✓ |
| 07 | Shrinkage | X | VF-E13 | N/A | ✓ |
| 08 | Blooming | (X) | VF-E17 | SATRA TM361 | ✓ |
| 09 | Hydrolysis | X | VF-E19 | SATRA TM344 | ✓ |
| 10 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 11 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 12 | Contact Discoloration | (X) | VF-E05 | N/A | ✓ |
| 13 | Hardness | X | VF-A01 | N/A | ✓ |
| 14 | Specific Gravity | X | VF-A04 | ASTM D792-20 | ✓ |
| 15 | Laminar Peel Strength | (X) | VF-F04 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.19 Zipper Performance Package Test for Full Test and Additional Color (Specification 4.22)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Chain Crosswise strength | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 02 | Element Pull-Off | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 03 | Element slippage, lengthwise | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 04 | Top Stop Holding Strength | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 05 | Bottom Stop Holding Strength, Slider | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 06 | Bottom Stop Holding Strength, Crosswise | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 07 | Bottom Stop Holding, Stringer Separation | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 08 | Bridge Top Stop, Stringer Separation | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 09 | Separable Pull off strength | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 10 | Fixed Retainer Pull off | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 11 | Separating Unit, Crosswise | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 12 | Resistance to twist of Pull and slider | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 13 | Resistance to pull -Off slider -pull | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |

©2026 VF Corporation. All Rights Reserved
Internal Use

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 14 | Resistance to angular-pull-of slider pull | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 15 | Holding strength of slider lock | X | | VF-B17 | ASTM D2061-07(2021) | ✓ |
| 16 | Slide Fasteners to Repeated Opening and Closing | X | | VF-B26 | SATRA TM50 | ✓ |
| 17 | Seam Stitch tear | (X) | | VF-B19 | SATRA TM180 | ✓ |
| 18 | Heat Aging | (X) | (X) | VF-E04 | N/A | ✓ |
| 19 | Shrinkage | X | | VF- E13 | N/A | ✓ |
| 20 | Color Fastness to Water | (X) | (X) | VF-E01 | N/A | ✓ |
| 21 | Color Fastness to Crocking | (X) | (X) | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 22 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 23 | Water Wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 24 | Corrosion Test | (X) | | VF- E09 | N/A | ✓ |
| 25 | pH Value (For zipper tape) | X | | ISO 3071 | ISO 3071 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

## 3.20 Straps & Plugs Performance Package Test for Full Test (Specification 4.23)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile | X | VF-B01 | N/A | ✓ |
| 02 | Strap Pull Strength | X | VF-F06 | N/A | ✓ |
| 03 | Ross Flex | (X) | VF-D02 | N/A | ✓ |
| 04 | Color Fastness to Crocking | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 05 | Color Fastness to Water | (X) | VF-E01 | N/A | ✓ |
| 06 | Color Fastness to Perspiration (for direct skin contact only) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 07 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 08 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 09 | Hydrolysis | (X) | SATRA TM344 | SATRA TM344 | ✓ |
| 10 | Shrinkage | X | VF-E13 | N/A | ✓ |
| 11 | Blooming | (X) | SATRA TM361 | SATRA TM361 | ✓ |
| 12 | Hardness | X | VF-A01 | N/A | ✓ |
| 13 | Specific Gravity | X | VF-A04 | ASTM D792-20 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.21 Ink & Printed Material Performance Package Test for Full Test (Specification 4.24)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Bally Flex | (X) | VF-D01 | ASTM D6182-23 | ✓ |
| 02 | ROSS Flex | (X) | VF-D02 | N/A | ✓ |
| 03 | Martindale Abrasion | (X) | VF-C04 | N/A | ✓ |

©2026 VF Corporation. All Rights Reserved
Internal Use

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 04 | Color Fastness to Crocking | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 05 | Color Fastness to Water | (X) | VF-E01 | N/A | ✓ |
| 06 | Color Fastness to Perspiration (for direct skin contact only) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 07 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 08 | Heat aging | X | VF-E04 | N/A | ✓ |
| 09 | Hydrolysis | X | SATRA TM344 | SATRA TM344 | ✓ |
| 10 | Ink Adhesion on Substrate | X | VF-F05 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.22 Insole Performance Package Test for Full Test (Specification 4.25)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Tensile | X | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 02 | Elongation | (X) | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 03 | Trouser Tear | (X) | VF-B07 | ASTM D2261-13(2024) | ✓ |
| 04 | Seam Stitch Tear | (X) | VF-B19 | SATRA TM180 | |
| 05 | Bursting | (X) | VF-B02 | ASTM D3786/D3786M-18(2023) | ✓ |
| 06 | Ross Flex | (X) | VF-D02 | N/A | ✓ |
| 07 | Laminar Peel Strength | (X) | VF-F04 | N/A | ✓ |
| 08 | Shrinkage | (X) | VF-E13 | N/A | ✓ |
| 09 | Static Water Absorption | (X) | VF-E10 | N/A | ✓ |
| 10 | Water Swelling | (X) | VF-E18 | N/A | ✓ |
| 11 | pH Value (for stroble insole) | X | ISO 3071 | ISO 3071 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.23 Elastic Cord Performance Package Test for Full Test and Additional Color (Specification 4.26)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D6775-13(2024) | ✓ |
| 02 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 03 | Limit of Useful Extension (%) | X | | SATRA TM102 | SATRA TM102 | ✓ |
| 04 | Resistance to Repeated Extension (10,000 cycles) | X | | SATRA TM103 | SATRA TM103 | ✓ |
| 05 | Resistance to Repeated Extension after heat ageing (10,000 cycles) | X | | ISO 20870 followed by TM 103 | ISO 20870 followed by TM 103 | ✓ |
| 06 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 07 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |

©2026 VF Corporation. All Rights Reserved
Internal Use

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 08 | Color Fastness to Perspiration (for direct skin contact only) | (X) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 09 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 10 | Heat aging | (X) | (X) | VF-E04 | N/A | ✓ |
| 11 | Hydrolysis | (X) | (X) | SATRA TM344 | SATRA TM344 | ✓ |
| 12 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 13 | pH Value | X | | ISO 3071 | ISO 3071 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

## 3.24 Wrapping Paper Performance Package Test for Full Test (Specification 4.27)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Color Fastness to Crocking | (X) | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 02 | PVC Migration Fastness | (X) | VF-E06 | ISO 15701-2022 | ✓ |
| 03 | Contact Discoloration | (X) | VF-E05 | N/A | ✓ |
| 04 | Heat aging | (X) | VF-E04 | N/A | ✓ |
| 05 | Hydrolysis | (X) | VF-E19 | SATRA TM344 | ✓ |
| 06 | Ink Adhesion on Substrate | X | VF-F05 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.25 Shoe Box Performance Package Test for Full Test (Specification 4.28)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Bursting | X | VF-B02 | N/A | ✓ |
| 02 | Color Fastness to Crocking | (X) | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 03 | Rubbing Fastness | (X) | VF-E24 (GB) | N/A | ✓ |
| 04 | Contact Discoloration | (X) | VF-E05 | N/A | ✓ |
| 05 | Compression | X | ASTM D642 | ASTM D642 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.26 Carton Performance Package Test for Full Test (Specification 4.28)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Bursting | X | VF-B02 | N/A | ✓ |
| 02 | Edge Crush Test | X | VF-B27 | SATRA TM440 | ✓ |
| 03 | Adhesion Strength: Dry Test Method | X | VF-F11 | N/A | ✓ |
| 04 | Adhesion Strength: Wet Test Method | X | VF-F11 | N/A | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

## 3.27 Woven Label Performance Package Test for Full Test and Additional Color (Specification 4.29)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Stitch Tear | X | | VF-B08 | SATRA TM33 | ✓ |
| 02 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 03 | Color Fastness to Water | X | X | VF-E01 | N/A | ✓ |
| 04 | Color Fastness to Perspiration (for direct skin contact only) | (X) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 05 | Color Fastness to Light (for light and neo color) | X | X | VF-E03 | N/A | ✓ |
| 06 | Heat aging | (X) | (X) | VF-E04 | N/A | ✓ |
| 07 | Hydrolysis (For printed/coated material) | (X) | (X) | VF-E19 | SATRA TM344 | ✓ |
| 08 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 09 | pH Value | X | | ISO 3071 | ISO 3071 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

## 3.28 Reinforcement Performance Package Test for Full Test and Additional Color (Specification 4.30)

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 01 | Tensile Strength | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 02 | Elongation | X | | VF-B01 | ASTM D5035-11(2024) | ✓ |
| 03 | Trouser Tear | X | | VF-B07 | ASTM D2261-13(2024) | ✓ |
| 04 | Stitch Tear | (X) | | VF-B08 | SATRA TM33 | ✓ |
| 05 | Bursting | X | | VF-B02 | ASTM D3786/D3786M-18(2023) | |
| 06 | Color Fastness to Crocking | X | X | VF-E02 | AATCC TM8-2016e(2022)e | ✓ |
| 07 | Color Fastness to Water | (X) | (X) | VF-E01 | N/A | ✓ |
| 08 | Color Fastness to Perspiration (for direct skin contact only) | (X) | (X) | VF-E16 | AATCC TM15-2021 | ✓ |
| 09 | Color Fastness to Light (for light and neo color) | (X) | (X) | VF-E03 | N/A | ✓ |
| 10 | Martindale Abrasion (for direct skin contact only) | (X) | (X) | VF-E04 | N/A | ✓ |
| 11 | Water wicking | (X) | (X) | VF-E14 | N/A | ✓ |
| 12 | pH Value | X | | ISO 3071 | ISO 3071 | ✓ |
| 13 | Yarn Count | (X) | | VF-A07 | ASTM D3775-17(2023) | ✓ |
| 14 | Weight | (X) | | VF-A02 | ASTM D3776/D3776M-20(2025) | ✓ |

©2026 VF Corporation. All Rights Reserved
Internal Use

| No. | Mandatory Test Item | Full Package | Additional Color | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|---|
| 15 | Flammability test | X | | CAN/CGSB 4.2 No. 27.5 | CAN/CGSB 4.2 No. 27.5 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | | |

## 3.29 Shoe Charms Performance Package Test for Full Test (Specification 4.31)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Pull Strength | X | VF-B04 | SATRA TM149 | ✓ |
| 02 | Hydrolysis | X | SATRA TM344 | SATRA TM344 | ✓ |
| 03 | Heat Aging | X | VF-E04 | N/A | ✓ |
| 04 | Washing Test | X | VF-G03 | N/A | ✓ |
| 05 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 06 | Corrosion Test | X | VF-E09 | N/A | ✓ |
| 07 | Security of attachment of stones or inlays | (X) | | SATRA TM275 | ✓ |
| 08 | Sharp Points | (X) | | CFR 1500.48 | ✓ |
| 09 | Sharp Edges | (X) | | CFR 1500.49 | ✓ |
| 10 | Small Parts | (X) | | CFR 1501 | ✓ |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

## 3.30 Whole Shoe Performance Package Test for Full Test (Specification 4.32, 4.33, 4.34 & 4.35&4.36)

| No. | Mandatory Test Item | Full Package | VF Test Method | International Test Method | Vans |
|---|---|---|---|---|---|
| 01 | Bonding Test | X | VF-F01 | N/A | ✓ |
| 02 | Strength of Hardware Attachment | (X) | VF-B04 | SATRA TM149 | ✓ |
| 03 | Strap Pull Strength | (X) | VF-F06 | N/A | ✓ |
| 04 | Whole Shoe Flex at Room Temperature | X | VF-G02 | N/A | ✓ |
| 05 | Whole Shoe Flex at Freezing | (X) | VF-G02 | N/A | ✓ |
| 06 | Whole Shoe Flex – Waterproof | (X) | VF-G04 | N/A | ✓ |
| 07 | Whole Shoe Waterproof – Static | (X) | VF-G07 | NA | ✓ |
| 08 | Whole Shoe Water Absorption | (X) | VF-G06 | NA | ✓ |
| 09 | Flexing Resistance Footwear / Outsole (GB) | (X) | VF-G05 | N/A | ✓ |
| 10 | Whole Shoe Wash Test (For shoes intended to be washed only) | (X) | VF-G03 | N/A | ✓ |
| 11 | Hydrolysis | X | VF-E19 | SATRA TM344 | ✓ |
| 12 | Heat Aging | X | VF-E04 | N/A | ✓ |
| 13 | Color Fastness to Light (for light and neo color) | X | VF-E03 | N/A | ✓ |
| 14 | Slip Resistance | (X) | VF-B25 | SATRA TM144 | |
| **Restricted Substance List (RSL) Test – Refer to RSL Implementation Manual** | | | | | |

©2026 VF Corporation. All Rights Reserved
Internal Use

# Section 4: Testing Requirement

## 4.1 Fabric Upper Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | | Material Specification Levels – Minimum Requirement | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength [D] | Tensile | VF-B01 | ASTM D5035-11(2024) | kgf/cm | Vamp/Quarter | All Directions | Thickness ≤1mm | 4.0 | 5.0 | 5.0 | 6.0 | 7.0 | 7.0 |
| | | | | | | | Thickness >1mm | 5.0 | 6.0 | 6.0 | 7.0 | 8.0 | 8.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 3.0 | 4.0 | 5.0 | 5.0 | 5.0 | 5.0 |
| | Elongation | VF-B01 | ASTM D5035-11(2024) | % | ALL Parts | All Directions | All Thickness | ≥10 | ≥10 | ≥10 | ≥10 | ≥10 | ≥10 |
| | Trouser Tear | VF-B07 | ASTM D2261-13(2024) | kgf | Vamp/Quarter | All Directions | Thickness ≤1mm | 1.5 | 1.5 | 1.5 | 2.0 | 3.0 | 3.0 |
| | | | | | | | Thickness >1mm | 2.0 | 2.0 | 2.0 | 3.0 | 4.0 | 4.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 1.5 | 1.5 | 1.5 | 2.0 | 2.0 | 2.0 |
| | Stitch tear | VF-B08 | SATRA TM 33 | kgf/cm | Vamp/Quarter | All Directions | Thickness ≤1mm | 2.0 | 2.0 | 3.0 | 3.0 | 4.0 | 5.0 |
| | | | | | | | Thickness >1mm | 3.0 | 3.0 | 4.0 | 4.0 | 6.0 | 6.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 2.0 | 2.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | Bursting [C] | VF-B02 | ASTM D3786/D3786M-18(2023) | $kgf/cm^2$ | Vamp/Quarter | All Directions | Thickness ≤1mm | 8 | 8 | 10 | 10 | 12 | 14 |
| | | | | | | | Thickness >1mm | 10 | 10 | 12 | 12 | 14 | 14 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 7.0 | 7.0 | 8.0 | 8.0 | 8.0 | 8.0 |
| Abrasion | Martindale [L] | VF-C04 | N/A | cycles | Vamp/Quarter /Other components [F] | Dry | All Thickness | 12.5K | 25K | 25K | 25K | 50K | 50K |
| | | | | | | Wet | All Thickness | 3.5K | 6.5K | 6.5K | 6.5K | 13K | 13K |
| Color fastness | Water [GP] | VF-E01 | N/A | AATCC scale | ALL Parts | 38 ±1 °C, 6 hrs. | All colors | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Crocking [GP] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC scale | ALL Parts | Dry | All colors | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | ALL Parts | Wet | All colors | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | Perspiration [GP] | VF-E16 | AATCC TM15-2021 | AATCC scale | ALL Parts | 38 ±1 °C, 6 hrs. | All colors | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | UV- Light [A] | VF-E03 | N/A | AATCC scale | ALL Parts | 24 hrs.,50 C | All Thickness | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV- Light [I] | VF-E03 | N/A | AATCC scale | ALL Parts | 12 hrs.,50 C | All Thickness | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Water Contact | Wicking | VF-E14 | N/A | mm | ALL Parts | Non-wicking/2 hours | All Thickness | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 |
| | Maeser [K] | VF-E11 | N/A | cycles | ALL Parts | 23 ±2 °C | All Thickness | 15K | 15K | 15K | 15K | 15K | 15K |
| | Dynamic Absorption [K] | VF-E11 | N/A | % | ALL Parts | 23 ±2 °C | All Thickness | <25 | <25 | <25 | <25 | <25 | <25 |
| | Static Water Absorption [K] | VF-E10 | N/A | % | ALL Parts | 23 ±2 °C | All Thickness | ≤ 50 | ≤ 50 | ≤ 50 | ≤ 50 | ≤ 50 | ≤ 50 |
| | Spray Test [E] | VF-E27 | ISO 4920 | rating | ALL Parts | 23 ±2 °C | All Thickness | 3 | 3 | 3 | 3 | 3 | 3 |
| Heat Resistance | Heat Aging (white & light color and coated & printed material) [B] | VF-E04 | N/A | AATCC scale | ALL Parts | 72 hrs., 70℃ | All Thickness | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Hydrolysis (white & light color and coated & printed material) [B] | VF-E19 | SATRA TM344 | AATCC scale | ALL Parts | 7 days /70°C /95% RH | All Thickness | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | rating | | | | No delamination of coating or ink | | | | | |
| Adhesive Strength | Peel Strength to Rubber | VF-F02 | N/A | kgf/cm | Vamp / Quarter **ONLY** | 23 ±2 °C | All Thickness | 3.0 | 3.0 | 3.0 | 3.0 | 3.5 | 3.5 |
| | Laminar Peel Strength | VF-F04 | | | ALL Parts | 23 ±2 °C | All Thickness | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 |
| | Finish Adhesion [B] | VF-F03 | N/A | kgf/cm | ALL Parts | Dry | All Thickness | 0.4 | 0.4 | 0.4 | 0.4 | 0.8 | 0.8 |
| | | | | | | Wet | All Thickness | 0.2 | 0.2 | 0.2 | 0.2 | 0.6 | 0.6 |
| | | VF-E19 | SATRA TM344 | kgf/cm | ALL Parts | Hydrolysis 7day/70°C /95% RH | All Thickness | 0.7 | 0.7 | 0.7 | 1.0 | 1.2 | 1.2 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Fabric Density | Yarn Count | VF-A07 | ASTM D3775-17(2023) | # of yarns | ALL Parts | Length | All Thickness | For woven fabric, refer to material specification per fabric | | | | | |
| | | | | | ALL Parts | Width | All Thickness | | | | | | |
| | Weight | VF-A02 | ASTM D3776/D3776M-20(2025) | g/m$^2$ | ALL Parts | 23 ±2 °C | All Thickness | Refer to material specification per fabric | | | | | |
| | pH Value [H] | | ISO 3071 | | ALL Parts | 23 ±2 °C | All Thickness | 4.0-7.5 | | | | | |
| | Flammability test [J] | | CAN/CGSB 4.2 No. 27.5 | | ALL Parts | 23 ±2 °C | All Thickness | >4seconds | | | | | |

**Notes:**

A- UV light test for light colors

**B-** Coated / laminated material

C- Material bursting is acceptable below specs if it stretches 1.0 cm or more above the top clamping ring without breaking or bursting

D- Fabric with lower material strength can be accepted if it is laminated with other material and the final strength of the combined material meets the specification.

E- For water resistance material

F- Other components could be tongue, side stripe, binding, or any part other than vamp and quarter

H- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

I- For neon color only

J- Flammability test MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

K- For waterproof coating or membrane material only

L- For printed upper fabric only

GP **– Guidelines on Proper use:**

>>> Crocking and colorfastness – any material which have been approved at levels below 3.0 should not be used in contact with light colored materials or materials with a significant contrast in color.

>>> Contact Discoloration – any combination of contacting materials (including laminating and tacking adhesives) having potential color migration issues should be tested using VF Test method. VF- E05" contact discoloration.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.2 Fabric Lining Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | | Material Specification Levels – Minimum Requirement | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength [E] | Tensile | VF-B01 | ASTM D5035-11(2024) | kgf/cm | Vamp / Quarter | All Directions | Thickness ≤ 1mm | 2.0 | 2.0 | 2.5 | 3.0 | 4.0 | 5.0 |
| | | | | | | | Thickness >1 mm | 2.5 | 3.0 | 3.0 | 4.0 | 6.0 | 7.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 2.0 | 2.0 | 2.0 | 3.0 | 3.0 | 3.0 |
| | Elongation | VF-B01 | ASTM D5035-11(2024) | % | ALL Parts | All Directions | All Thickness | ≥10 | ≥10 | ≥10 | ≥10 | ≥10 | ≥10 |
| | Trouser Tear | VF-B07 | ASTM D2261-13(2024) | kgf | Vamp/ Quarter | All Directions | Thickness ≤ 1mm | 1.5 | 1.5 | 1.5 | 1.5 | 2.0 | 2.5 |
| | | | | | | | Thickness >1 mm | 2.0 | 2.0 | 2.0 | 2.0 | 3.0 | 3.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 1.5 | 1.5 | 1.5 | 2.0 | 2.0 | 2.0 |
| | Stitch Tear [A] | VF-B08 | SATRA TM 33 | kgf/cm | Vamp / Quarter | All Directions | Thickness ≤1mm | 2.0 | 2.0 | 3.0 | 3.0 | 4.0 | 4.0 |
| | | | | | | | Thickness >1 mm | 2.5 | 2.5 | 4.0 | 4.0 | 6.0 | 6.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 2.0 | 2.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | Bursting [B] | VF-B02 | ASTM D3786/D3786M-18(2023) | $kgf/cm^2$ | Vamp / Quarter | All Directions | Thickness ≤ 1mm | 8.0 | 8.0 | 10.0 | 10.0 | 11.0 | 12.0 |
| | | | | | | | Thickness >1 mm | 10.0 | 10.0 | 12.0 | 12.0 | 14.0 | 14.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | All Thickness | 7.0 | 7.0 | 8.0 | 8.0 | 8.0 | 8.0 |
| Abrasion | Martindale | VF-C04 | N/A | cycles | Vamp / Quarter | Dry | All Thickness | 12.5K | 25K | 25K | 25K | 50K | 50K |
| | | | | | | Wet | All Thickness | 3.5K | 6.5K | 6.5K | 6.5K | 13K | 13K |
| | | | | | Sandal Strap and other component [F] | Dry / Wet | All colors | All values: ≥ 50% of above specs | | | | | |
| Color fastness | Crocking [GP] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC Scale | ALL Parts | Dry | All colors | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | | Wet | All colors | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | Perspiration | VF-E16 | AATCC TM15-2021 | | ALL Parts | 38±1°C,6 hrs. | All colors | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [C] | VF-E03 | N/A | | ALL Parts | 24 hrs,50°C | All Thickness | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | | Material Specification Levels – Minimum Requirement | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | UV-light [D] | VF-E03 | N/A | | ALL Parts | 12 hrs,50°C | All Thickness | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | ALL Parts | 72 hrs., 70 °C | All Thickness | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| Adhesive Strength | Laminar Peel Strength | VF-F04 | N/A | kgf/cm | ALL Parts | 23 ±2 °C | All Thickness | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| Water Contact / **Water Management** | Wicking | VF-E14 | N/A | mm | ALL Parts | Non-wicking/2 hrs | All Thickness | ≤10 | ≤10 | ≤10 | ≤10 | ≤10 | ≤10 |
| | | | | | | Material designed to wick /1hour | All Thickness | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 |
| | Static Water Absorption | VF-E10 | N/A | % | ALL Parts | 23 ±2 °C | All Thickness | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 |
| | Spray Test | VF-E27 | ISO 4920 | rating | ALL Parts | 23 ±2 °C | All Thickness | 3 | 3 | 3 | 3 | 3 | 3 |
| Fabric Density | Yarn Count | VF-A07 | ASTM D3775-17(2013) | # of yarns | ALL Parts | Length | All Thickness | For woven fabric, refer to material specification per material | | | | | |
| | | | | | ALL Parts | Width | All Thickness | | | | | | |
| | Weight | VF-A02 | ASTM D3776/D3776M-20(2025) | $g/m^2$ | ALL Parts | 23 ±2 °C | All Thickness | Refer to material specification per material | | | | | |
| | pH Value [H] | | ISO 3071 | | ALL Parts | 23 ±2 °C | All Thickness | 4.0-7.5 | | | | | |
| | Flammability test [I] | | CAN/CGSB 4.2 No. 27.5 | | ALL Parts | 23 ±2 °C | All Thickness | >4seconds | | | | | |

**Notes #1**

A- Stitch Tear for Woven fabric = 2.0 kgf/cm

B- Material bursting is acceptable below specs if it stretches 1.0 cm or more above the top clamping ring without breaking or bursting

C- For light colors

D- For neon colors

E- For fabric with lower material strength could still be used for lining provided it will be laminated with upper. The physical properties will be checked again with laminated material.

**F- Other component could be tongue or any part other than vamp and quarter**

H- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

I- Flammability test MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

GP **– Guidelines on Proper use:**

>>> Crocking and colorfastness – any material which have been approved at levels below 3.5 Dry and 3.0 Wet should not be used in contact with light colored materials or materials with a significant contrast in color.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.3 Leather Upper Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength [I] | Tensile | VF-B01 | N/A | kgf/cm | Vamp / Quarter | ALL Directions | 16 | 20 | 20 | 20 | 25 | 25 |
| | | | | | Sandal Strap and other component [M] | ALL Directions | 10 | 12 | 12 | 14 | 14 | 14 |
| | Elongation | VF-B01 | N/A | % | ALL Parts | ALL Directions | ≥25 | ≥25 | ≥25 | ≥25 | ≥25 | ≥25 |
| | Baumann Tear | VF-B09 | ISO 3377-2 | kgf | Vamp / Quarter | ALL Directions | 4 | 6 | 6 | 8 | 8 | 10 |
| | | | | | Sandal Strap and other component [M] | ALL Directions | 3 | 4 | 4 | 6 | 14 | 14 |
| | Stitch tear | VF-B08 | SATRA TM 33 | kgf/cm | Vamp / Quarter | ALL Directions | 3 | 3.5 | 3.5 | 4 | 5 | 5 |
| | | | | | Sandal Strap and other component [M] | ALL Directions | 2 | 2 | 3 | 3 | 3 | 3 |
| | Bursting [H] | VF-B02 | N/A | $kgf/cm^2$ | Vamp / Quarter | ALL Directions | 14 | 16 | 16 | 16 | 18 | 18 |
| | | | | | Sandal Strap and other component [M] | ALL Directions | 10 | 10 | 12 | 12 | 12 | 12 |
| Abrasion | Taber [A] | VF-C05 | ASTM D7255-22e1(2022) | cycles | Vamp / Quarter | H-22 / 1000g | 100 | 150 | 150 | 150 | 200 | 200 |
| | Taber [B] | | | cycles | | H-22 / 1000g | 50 | 50 | 50 | 50 | 75 | 75 |
| | Taber [C] | | | cycles | | CS-17 / 500g | 500 | 500 | 500 | 500 | 1000 | 1000 |
| | Taber -All types | | | cycles | Sandal Strap and other component [M] | All types of abrasive wheels | All values: ≥ 50% of above specs | | | | | |
| Flexing | Bally [D] | VF-D01 | ASTM D6182-23 | cycles | Vamp / Quarter | Dry | 100K | 100K | 100K | 100K | 125K | 125K |
| | | | | | | Wet | 20K | 20K | 20K | 20K | 25K | 25K |
| | | | | | Sandal Strap and other component [M] | Dry / Wet | All values: ≥ 50% of above specs | | | | | |
| | | | ASTM D6182-23, SATRA TM 344 | | ALL Parts | Hydrolysis, 7 days/60°C /95% RH | 50K | 50K | 50K | 50K | 75K | 75K |
| | | | | | ALL Parts | (- 20 °C) [A] | 30K | 30K | 30K | 30K | 50K | 50K |
| Color fastness | Water [GP, G] | VF-E01 | N/A | AATCC scale | ALL Parts | PU Coated, Full Grain | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | | Suede, Nubuck, Aniline | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | Crocking GP, G | VF-E02 | ISO 20433: 2024(E) | | ALL Parts | PU Coated, Full Grain – DRY | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | | PU Coated, Full Grain – WET | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | | | | | | suede, Nubuck, Aniline -DRY | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 |
| | | | | | | Suede, Nubuck, Aniline – WET | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | Perspiration GP, G | VF-E16 | AATCC TM15-2021 | | ALL Parts | PU Coated, Full Grain | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | ALL Parts | Suede, Nubuck, Aniline | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | UV- Light F | VF-E03 | N/A | | ALL Parts | 24 hrs.,50°C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV- Light T | VF-E03 | N/A | | ALL Parts | 12 hrs.,50°C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | PVC Migration | VF-E06 | ISO 15701-2022 | | ALL Parts | 16 hrs.,50°C | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | ALL Parts | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Hydrolysis A | VF-E19 | SATRA TM344 | AATCC scale | ALL Parts | 7 days/70°C /95% RH | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | rating | ALL Parts | | No delamination of coating | | | | | |
| Adhesive strength | Peel Strength to Rubber | VF-F02 | N/A | kgf/cm | Vamp / Quarter **ONLY** | 23 ±2 °C | 3 | 3 | 3 | 3 | 3.5 | 3.5 |
| | Laminar Peel Strength | VF-F04 | | | ALL Parts | | 0.4 | 0.4 | 0.4 | 0.4 | 0.8 | 0.8 |
| Finish Adhesion | Finish Adhesion | VF-F03 | N/A | kgf/cm | ALL Parts A | Dry | 2.0 | 2.0 | 2.0 | 2.0 | 2.5 | 2.5 |
| | | | | | | Wet | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| | | | | | ALL Parts | Dry: Full grain *( With waxes & finishes on top)* | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | | VF-E19 | SATRA TM344 | kgf/cm | ALL Parts [A] | Hydrolysis, 7 days/70°C /95% RH | 1.2 | 1.2 | 1.2 | 1.2 | 1.2 | 1.2 |
| | Tape Test | VF -F05 | N/A | rating | ALL Parts | 23 ±2 °C | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| Water Contact | Maeser [P] | VF-E11 | N/A | cycles | ALL Parts | 23 ±2 °C | 15K | 15K | 15K | 15K | 15K | 15K |
| | Dynamic [P] Absorption | | N/A | % | ALL Parts | 23 ±2 °C | <20 | <20 | <20 | <20 | <20 | <20 |
| | Static Water Absorption [P] | VF-E10 | ASTM D6015-21(2022) | % | ALL Parts | 23 ±2 °C | ≤ 30 | ≤ 30 | ≤ 30 | ≤ 30 | ≤ 30 | ≤ 30 |
| | Wicking | VF-E14 | N/A | mm | ALL Parts | 2 hours | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 |
| | pH Value [S] | | ISO 4045 | | ALL Parts | 23 ±2 °C | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 |
| Extractable | Oil Content [K] | VF-I01 | N/A | % | ALL Parts | 23 ±2 °C | <18 | <18 | <18 | <18 | <18 | <18 |

**Notes #1**

**A-** Polyurethane coated leather including split leather PU coated

**B-** Full grain leather and Naked leather **Note: Full Grain leather, no need to test.**

**C-** Embossed leather & tumbled soft leather

D - Evaluation: Pass - Rank 2 Discoloration or thin cracks in top finish coat only

**F-** UV light test is for light colors, For Egypt market, requirement is Min 5 Grade

G- For Egypt market, requirement is Min 3 Grade

**H-** Material bursting is acceptable below specs requirement if it stretches 1.0 cm or more above the top clamping ring without breaking or bursting

**I-** For leather with lower material strength can still be used for upper if it has backer. The physical properties will be checked again with laminated backer

**K-** Oiled leather, for special case, please refer to individual specification

**L-** For leather with perforation or holes and 2 different upper materials combined by stitching.

**M-** Other components: could be tongue, side stripe, binding, or any part as determined by Brand Developer

**P** – To be performed for waterproof material only

**S-** pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

**T-** For neon colors

**GP – Guidelines on Proper use:**

- Crocking and colorfastness – any material which have been approved at levels below 3.5 for dry and 3.0 for wet should not be used in contact with light colored materials or materials with a significant contrast in color.

- Contact Discoloration – any combination of contacting materials (including laminating and tacking adhesives) having potential color migration issues should be tested using VF Test method. VF-E05" contact discoloration "

**Notes #2**

1. Use of a leather that does not meet the VF material specification is allowed provided:
   - Leather meets the tannage specification provided and approved by GMS Team
2. For THICKNESS and WEIGHT requirement REFER TO INDIVIDUAL TANNAGE SPECIFICATION SHEET DATA

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.4 Leather Lining Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength [A] | Tensile | VF-B01 | N/A | kgf/cm | Vamp / Quarter | All Directions | 2.5 | 4.0 | 4.0 | 5.0 | 6.0 | 7.0 |
| | | | | | Sandal Strap and other component [G] | All Directions | 2 | 3.0 | 3.0 | 5.0 | 5.0 | 5.0 |
| | Elongation | VF-B01 | N/A | % | ALL Parts | All Directions | ≥ 15 | ≥ 15 | ≥ 15 | ≥ 15 | ≥ 15 | ≥ 15 |
| | Baumann Tear | VF-B09 | ISO 3377-2 | kgf | Vamp / Quarter | All Directions | 2.5 | 3.5 | 3.5 | 3.5 | 4.5 | 4.5 |
| | | | | | Sandal Strap and other component [G] | All Directions | 2 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | Stitch Tear | VF-B08 | SATRA TM33 | kgf/cm | Vamp / Quarter | All Directions | 3 | 3.5 | 3.5 | 3.5 | 4.0 | 4.0 |
| | | | | | Sandal Strap and other component [G] | All Directions | 2 | 2.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | Bursting [D] | VF-B02 | N/A | $kgf/cm^2$ | Vamp / Quarter | All Directions | 8 | 10 | 10 | 10 | 12 | 12 |
| | | | | | Sandal Strap and other component [G] | All Directions | 7 | 8 | 8 | 8 | 8 | 8 |
| Abrasion | Martindale [C] | VF-C04 | N/A | cycles | Vamp / Quarter | Dry | 12.5K | 25K | 25K | 25K | 50.0 | 50K |
| | | | | | | Wet | 3.5K | 6.5K | 6.5K | 6.5K | 12.5K | 12.5K |
| | | | | | Sandal Strap and other component [G] | Dry / Wet | All values: ≥ 50% of above specs | | | | | |
| Flexing | Bally [H] | VF-D01 | ASTM D6182-23 | flexes | Vamp / Quarter | Dry | 100K | 100K | 100K | 100K | 100K | 125K |
| | | | | | | Wet | 20K | 20K | 20K | 20K | 20K | 25K |
| | | | | | Sandal Strap and other component [G] | Dry / Wet | All values: ≥ 50% of above specs | | | | | |
| | | | ASTM D6182-23, | | ALL Parts | Hydrolysis, 7 days / **70°C** / 95% RH [E] | 50K | 50 K | 50 K | 50 K | 75K | 75K |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | | | SATRA TM 344 | | | (- 20° C) [E] | 30K | 30K | 30K | 30K | 30K | 30K |
| Colorfastness | Crocking [GP] | VF-E02 | ISO 20433: 2024(E) | AATCC Scale | ALL Parts | PU Coated, Full Grain- DRY | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | | PU Coated, Full Grain – WET | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | | | | | | Suede, Nubuck, Aniline – DRY | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 |
| | | | | | | Suede, Nubuck, Aniline – WET | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | Perspiration [GP] | VF-E16 | AATCC TM15-2021 | | ALL Parts | PU Coated, Full Grain | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | | Suede, Nubuck, Aniline | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | UV-light [F] | VF-E03 | N/A | | ALL Parts | 24 hrs., 50°C | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [O] | VF-E03 | N/A | | ALL Parts | 12 hrs., 50°C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | PVC Migration | VF-E06 | ISO 15701-2022 | | ALL Parts | 16 hrs.,50°C | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | rating | ALL Parts | 72 hrs, 70° C | 4.0 | | | | | |
| | Hydrolysis [B] | VF-E19 | SATRA TM344 | | ALL Parts | 7 days / 70°C / 95% RH | No delamination of coating | | | | | |
| | | | | | | | 4.0 | | | | | |
| Adhesive Strength | Finish Adhesion [B] | VF-F03 | N/A | kgf/cm | ALL Parts | Dry | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | | | | | | Wet | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| Water Contact | Wicking | VF-E14 | N/A | mm | ALL Parts | Non-wicking/2hrs. | ≤10 | ≤10 | ≤10 | ≤10 | ≤10 | ≤10 |
| | | | | | | Material designed to wick/1hour | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 |
| | Static Water Absorption | VF-E10 | ASTM D6015-21(2022) | % | ALL Parts | 23 ±2 °C | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 |
| | pH Value [N] | | ISO 4045 | | ALL Parts | 23 ±2 °C | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 | 3.2-7.5 |

**Notes #1**

A- For leather with lower material strength could still be used for lining provided it will be laminated to upper. The physical properties will be checked again with laminated material.

B- Polyurethane coated leather including split leather PU coated

C- Pig skin/Lamp/Goat leather Martindale = Dry: 8K minimum / Wet no need

©2026 VF Corporation. All Rights Reserved



Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |

D- Material bursting is acceptable below specs if it stretches 1.0 cm or more above the top clamping ring without breaking or bursting
F- UV light test is for light colors. For Egypt market, requirement is Min 5 Grade.
G- Other components: could be tongue or any part as determined by Brand Developer
H - Evaluation: Pass - Rank 2 Discoloration or thin cracks in top finish coat only
N- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.
O- For neon color only
GP **– Guidelines on Proper use:**
Crocking and colorfastness – any material which have been approved at levels below 3.5 for dry and 3.0 for wet should not be used in contact with light colored materials or materials with a significant contrast in color.
**Notes #2**

1. Use of a leather that does not meet the VF material specification is allowed provided:
   - Leather meets the tannage specification provided and approved by GMS Team
2. For THICKNESS and WEIGHT requirement REFER TO INDIVIDUAL TANNAGE SPECIFICATION SHEET DATA

## 4.5 Synthetic Leather Upper Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength [H] | Tensile | VF-B01 | ASTM D5035-11(2024) | kgf/cm | Vamp / Quarter | All Directions | 7.0 | 8.0 | 8.0 | 10 | 12 | 14 |
| | | | | | Sandal Strap and other component [I] | All Directions | 5.0 | 6.0 | 6.0 | 8.0 | 8.0 | 8.0 |
| | Elongation | VF-B01 | ASTM D5035-11(2024) | % | ALL Parts | All Directions | ≥15 | ≥15 | ≥15 | ≥15 | ≥15 | ≥15 |
| | Trouser Tear | VF-B07 | ASTM D2261-13(2024) | kgf | Vamp / Quarter | All Directions | 3.0 | 4.0 | 4.0 | 4.0 | 5 | 6 |
| | | | | | Sandal Strap and other component [I] | All Directions | 2.0 | 2.0 | 2.0 | 2.5 | 2.5 | 2.5 |
| | Stitch tear | VF-B08 | SATRA TM33 | Kgf/cm | Vamp / Quarter | ALL Directions | 3.0 | 3.5 | 3.5 | 4.0 | 5 | 5 |
| | | | | | Sandal Strap and other component [I] | ALL Directions | 2.0 | 2.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| | Bursting [E] | VF-B02 | | $kgf/cm^2$ | Vamp / Quarter | All Directions | 10 | 10 | 12 | 12 | 14 | 14 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | | | ASTM D3786/D3786M-18(2023) | | Sandal Strap and other component [I] | All Directions | 7.0 | 7.0 | 7.0 | 8.0 | 8.0 | 8.0 |
| Abrasion | Taber | VF-C05 | ASTM D7255-22e1(2022) | cycles | Vamp / Quarter | CS-17/500g | 500 | 500 | 500 | 500 | 500 | 1000 |
| | | | | | Sandal Strap and other component [I] | CS-17/500g | 200 | 200 | 200 | 200 | 200 | 200 |
| Flexing | Bally [G] | VF-D01 | ASTM D6182-23 | cycles | Vamp / Quarter | Dry | 100K | 100K | 100K | 100K | 125K | 125K |
| | | | | | | Wet | 20K | 20K | 20K | 20K | 25K | 25K |
| | | | | | Sandal Strap and other component [I] | Dry | 40K | 40K | 40K | 40K | 40K | 40K |
| | | | | | | Wet | 10K | 10K | 10K | 10K | 10K | 10K |
| | | | ASTM D6182-23, SATRA TM 344 | | ALL Parts | Hydrolysis, 7 days/70°C/ 95% RH) [A] | 50 K | 50 K | 50 K | 50 K | 75K | 75K |
| | | | | | ALL Parts | (- 20 C) [A] | 30K | 30K | 30K | 30K | 50K | 50K |
| Color fastness | Water [B, D] | VF-E01 | N/A | AATCC scale | ALL Parts | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Crocking [C, D, GP] | VF-E02 | AATCC TM8-2016e(2022)e | | ALL Parts | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV- Light [K] | VF-E03 | N/A | | ALL Parts | 24 hrs., 50 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV- Light [L] | VF-E03 | N/A | | ALL Parts for neon color | 12 hrs., 50 °C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | ALL Parts | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Hydrolysis [A] | VF-E19 | SATRA TM344 | AATCC scale | ALL Parts | 7days/70°C/ 95%RH [A] | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | rating | | | No delamination of coating | | | | | |
| Adhesive Strength | Peel Strength to Rubber | VF- F02 | N/A | kgf/cm | Vamp / Quarter **ONLY** | 23 ±2 °C | 3 | 3 | 3 | 3 | 3.5 | 3.5 |
| | Laminar Peel Strength | VF-F04 | | | ALL Parts | 23 ±2 °C | 0.4 | 0.4 | 0.4 | 0.4 | 0.8 | 0.8 |
| | Finish Adhesion | VF-F03 | N/A | kgf/cm | Vamp / Quarter | Dry / Wet | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | | | | | Sandal Strap and other component [I] | Dry / Wet | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| | | VF-E19 | SATRA TM344 | | ALL Parts | 7 days/70°C/ 95% RH [A] | 1.2 | 1.2 | 1.2 | 1.2 | 1.2 | 1.2 |
| Waterproof | Maeser [F] | VF-E11 | N/A | cycles | ALL Parts | 23 ±2 °C | 15K | 15K | 15K | 15K | 15K | 15K |
| | Dynamic Absorption [F] | | | % | ALL Parts | 23 ±2 °C | < 25 | < 25 | < 25 | < 25 | < 25 | < 25 |
| | Static Water Absorption [F] | VF-E10 | N/A | % | ALL Parts | 23 ±2 °C | ≤ 50 | ≤ 50 | ≤ 50 | ≤ 50 | ≤ 50 | ≤ 50 |
| | Wicking | VF-E14 | N/A | mm | ALL Parts | Non-wicking/2 hours | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 |

**Notes:**

A- Polyurethane coated synthetic leather including synthetic split leather PU coated

B- Dark Colored synthetic suede and synthetic Nubuck require minimum 3.0

C- Dark Colored synthetic suede and synthetic Nubuck require minimum Dry 2.5, wet 2.0

D- Oiled synthetic leather will require OVERRIDE specs since their performance normally is between 1.5- 2.0 for colorfastness to water and crocking and will transfer color to any contact material. Refer to GP – guidelines on proper use

E- Material bursting is acceptable below specs requirement if it stretches 1.0 cm or more above the top clamping ring without breaking or bursting

F- To be performed for Waterproof material only

G - Evaluation: Pass - Rank 2 Discoloration or thin cracks in top finish coat only

H- For synthetic leather with lower material strength can still be used for upper if it has backer. The physical properties will be checked again with laminated backer

I- Other components could be tongue, side stripe, binding or any part other than vamp and quarter

K- UV light test is for light colors.

L- For neon color only

> For THICKNESS and WEIGHT refer to material specification per material

GP- **Guidelines on Proper use:**

> Crocking and colorfastness – any material which have been approved at levels below 4.0 for dry and 3.5 for wet should not be used in contact with light colored materials or materials with a significant contrast in color.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.6 Synthetic Leather Lining Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength [E] | Tensile | VF-B01 | ASTM D5035-11(2024) | kgf/cm | Vamp / Quarter | All Directions | 2.5 | 4.0 | 4.0 | 5.0 | 6.0 | 6.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | 2.0 | 3.0 | 3.0 | 4.0 | 4.0 | 4.0 |
| | Elongation | VF-B01 | ASTM D5035-11(2024) | % | ALL Parts | All Directions | ≥15 | ≥15 | ≥15 | ≥15 | ≥15 | ≥15 |
| | Trouser Tear | VF-B07 | ASTM D2261-13(2024) | kgf | Vamp / Quarter | All Directions | 2 | 3 | 3 | 4 | 4 | 5 |
| | | | | | Sandal Strap and other component [F] | All Directions | 1.5 | 2 | 2 | 3 | 3 | 3 |
| | Stitch Tear | VF-B08 | SATRA TM 33 | kgf/cm | Vamp / Quarter | All Directions | 3 | 3.5 | 3.5 | 4.0 | 4.0 | 4.5 |
| | | | | | Sandal Strap and other component [F] | All Directions | 2.0 | 2.0 | 2.5 | 3.0 | n/a | n/a |
| | Bursting [D] | VF-B02 | ASTM D3786/D3786M-18(2023) | $kgf/cm^2$ | Vamp / Quarter | All Directions | 8.0 | 10.0 | 12.0 | 12.0 | 14.0 | 14.0 |
| | | | | | Sandal Strap and other component [F] | All Directions | 5.0 | 6.0 | 6.0 | 7.0 | 7.0 | 7.0 |
| Abrasion | Martindale | VF-C04 | N/A | cycles | Vamp / Quarter/other component F | Dry | 12.5K | 25K | 25K | 25K | 50K | 50K |
| | | | | | | Wet | 3.5K | 6.5K | 6.5K | 6.5K | 12.5K | 12.5K |
| | | | | | Sandal Strap | Dry | 5K | 10K | 10K | 10K | 10K | 10K |
| | | | | | | Wet | 1.5K | 3.0K | 3.0K | 3.0K | 3.0K | 3.0K |
| Flexing | Bally [G] | VF-D01, VF-E19 | ASTM D6182-23 | flexes | Vamp / Quarter | Dry | 100K | 100K | 100K | 100K | 125K | 125K |
| | | | | | | Wet | 20K | 20K | 20K | 20K | 25K | 25K |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | Material Specification Levels – Minimum Requirement | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | | | | | Sandal Strap and other component [F] | Dry | 40K | 40K | 40K | 40K | 40K | 40K |
| | | | | | | Wet | 10K | 10K | 10K | 10K | 10K | 10K |
| | | | ASTM D6182-23, SATRA TM344 | | ALL Parts | 7 days / 70°C / 95% RH [A] | 50K | 50K | 50K | 50K | 75K | 75K |
| | | | | | ALL Parts | (- 20 °C) [A] | 30K | 30K | 30K | 30K | 30K | 30K |
| Colorfastness | Crocking [C, GP] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC scale | ALL Parts | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Perspiration [B] | VF-E16 | AATCC TM15-2021 | | ALL Parts | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [H] | VF-E03 | N/A | | ALL Parts | 24 hrs., 50° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV-light [I] | VF-E03 | N/A | | ALL Parts for neon color | 12 hrs., 50° C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | ALL Parts | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Hydrolysis | VF-E19 | SATRA TM344 | | ALL Parts | 7days/ 70°C / 95% RH [A] | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | rating | ALL Parts | | No delamination of coating | | | | | |
| Adhesive Strength | Finish Adhesion [A] | VF-F03 | N/A | kgf/cm | ALL Parts | Dry | 0.4 | 0.4 | 0.4 | 0.4 | 0.8 | 0.8 |
| | | | | | ALL Parts | Wet | 0.2 | 0.2 | 0.2 | 0.2 | 0.6 | 0.6 |
| Water Contact | Wicking | VF-E14 | N/A | mm | ALL Parts | Non-wicking/2hours | ≤10 | ≤10 | ≤10 | ≤10 | ≤10 | ≤10 |
| | | | | | ALL Parts | material designed to wick/1hour | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 | ≥ 101.6 |
| | Static Water Absorption | VF-E10 | N/A | % | ALL Parts | 23 ±2 °C | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 | ≥ 10 |

**Notes #1**

A-Polyurethane – coated synthetic.

B- Dark Colored synthetic suede and synthetic Nubuck require minimum 3.0

C- Dark Colored synthetic suede and synthetic Nubuck require minimum Dry 2.5, Wet 2.0, but for collar and heel lining should be not lower than 3.0.

D- Material bursting is acceptable below specs requirement if it stretches 1.0 cm or more above the top clamping ring without breaking or bursting

E- For synthetic leather with lower material strength could still be used for lining provided it will be laminated to upper. The physical properties will be checked again with laminated material.

F- Other component could be tongue or any part other than vamp and quarter

G - Evaluation: Pass - Rank 2 Discoloration or thin cracks in top finish coat only

H- UV light test is for light colors.

©2026 VF Corporation. All Rights Reserved



Internal Use

I- For neon color only.
> For THICKNESS and WEIGHT refer to material specification per material
GP- **Guidelines on Proper use:**
>Crocking and colorfastness – any material which have been approved at levels below 4.0 for dry and 3.5 for wet should not be used in contact with light colored materials or materials with a significant contrast in color.

>Contact Discoloration – any combination of contacting materials (including laminating and tacking adhesives) having potential color migration issues should be tested using VF Test method. VF-E05 contact discoloration

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.7 Hardware Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Heat Resistance | Hydrolysis | VF-E19 | SATRA TM344 | % | 7days, 70°C 95 % RH | 80% retained of the original strength | | | | | |
| | | | | rating | | No peeling off coating and no deformation of material | | | | | |
| | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70 °C | 4.0 | | | | | |
| Colorfastness | UV- Light [A] | VF-E03 | N/A | | 24hrs, 50 °C | 4.0 | | | | | |
| | UV- Light [B] | VF-E03 | N/A | | 12hrs, 50 °C | 3.0 | | | | | |
| Chemical Resistance | Corrosion Test | VF- E09 | N/A | rating | 23±2°C / 48 hrs. | No Corrosion | | | | | |

**Note:**

A- UV light test is for light colors.
B- For neon color only.

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.8 Rubber Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels – *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile [A] | VF-B01 | N/A | kgf/cm$^2$ | 23 ±2 °C | 90 | 90 | 90 | 90 | 120 | 120 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 325 | 325 | 325 | 325 | 325 | 325 |
| | Die C Tear | VF-B05 | ASTM D624-00(2020) | kgf/cm | 23 ±2 °C | 30 | 30 | 30 | 35 | 40 | 40 |
| | Seam Stitch Tear- soling | VF-B20 | N/A | kgf/cm | 23 ±2 °C | 25 | 25 | 25 | 25 | 35 | 35 |
| Flexing | Ross Flex | VF-D02 | N/A | % growth | 23 ±2 °C /100K | < 50% | < 50% | < 50% | < 50% | < 50% | < 50% |
| | | | | | -15°C /100K | < 100% | < 100% | < 100% | < 100% | < 100% | < 100% |
| | | | | | -25°C /100K | -- | -- | -- | -- | -- | < 100% |
| Colorfastness | UV- Light [C] | VF-E03 | N/A | AATCC scale | 24 hrs., 50 C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV- Light [D] | VF-E03 | N/A | AATCC scale | 12 hrs., 50 C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Blooming Test [E] | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | | | |
| | Hydrolysis Test [E] | VF-E19 | SATRA TM344 | AATCC scale | 7days, 70 °C, 95% RH | 4.0 | | | | | |
| | | | | % | | 80% retained of the original strength | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 70 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Peel Strength to Upper | VF-F02 | N/A | kgf/cm | 23 ±2 °C | 2.5 | 2.5 | 3 | 3 | 3.5 | 3.5 |
| Hardness | Hardness | VF-A01 | N/A | A | 23 ±2 °C | Refer to specific requirement per type of Rubber Outsole | | | | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm$^3$ | 23 ±2 °C | Refer to specific requirement per type of Rubber Outsole | | | | | |
| Ground contact | DIN abrasion [B] | VF-C01 | N/A | mm$^3$ | 23 ±2 °C | < 250 | < 150 | < 150 | < 150 | <120 | <120 |
| | | VF-B25 | SATRA TM144 | COF | Dry * | ≥.0.40 | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥0.40 | ≥0.40 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | Coefficient of Friction -TM144 | | | | Wet * | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥0.30 |
| | Non- Marking | VF-E21 | N/A | rating | 23 ±2 °C | non- marking (5.0) | | | | | |

**Notes:**

**A**- Compound for cold weather applications should be ≥ 100 kgf/cm2

**B**- Compound for cold weather applications should be ≤150 mm3

**C**- UV light test is for light colors

**D**- For neon colors only

E - Blooming and Hydrolysis should be tested on sole unit

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> If specific gravity is ***beyond the standard range of ±*** 0.02, then the material should be rejected

>>> If Durometer hardness is ***beyond the standard range of*** ± 3, then the material should be rejected

* For slip resistance specification, a higher standard maybe required for specific purposes of use

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.9 Blown Rubber Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm$^2$ | 23 ±2 °C | 35 | 35 | 35 | 35 | 35 | 35 |
| | | | | | Sandal construction | 12 | 12 | 12 | 18 | n/a | n/a |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 250 | 250 | 250 | 250 | 250 | 250 |
| | | | | | Sandal construction | 175 | 175 | 175 | 200 | n/a | n/a |
| | Die C Tear | VF-B05 | ASTM D624-00(2020) | kgf/cm | 23 ±2 °C | 30 | 30 | 30 | 30 | 30 | 30 |
| | | | | | Sandal construction | 6 | 6 | 6 | 10 | n/a | n/a |
| | Split Tear | VF-B06 | N/A | kgf/cm | 23 ±2 °C | n/a | n/a | n/a | 2.5 | 2.5 | 2.5 |
| | | | | | Sandal construction | 1.5 | 1.5 | 1.5 | 2.0 | n/a | n/a |
| | Seam Stitch Tear -Soling | VF- B20 | N/A | kgf/cm | 23 ±2 °C | 25 | 25 | 25 | 25 | 25 | 25 |
| Compression Set | Static Compression Set | VF-B10 | N/A | % | 6 hrs., 45°C | ≤60.0 | ≤60.0 | ≤60.0 | ≤50.0 | ≤50.0 | ≤50.0 |
| Flexing | Ross Flex | VF-D02 | N/A | % growth | 23 ±2 °C /100K | < 50% | < 50% | < 50% | < 50% | < 50% | < 50% |
| | | | | | -15°C /100K | < 100% | < 100% | < 100% | < 100% | < 100% | < 100% |
| Colorfastness | UV- Light [A] | VF-E03 | N/A | AATCC scale | 24 hrs., 50 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | rating | 70 °C, 95% RH/7 days | No deterioration of physical appearance | | | | | |
| | | | | % | | 80% retained of the original strength | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 60 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Peel Strength to Upper | VF-F02 | N/A | kgf/cm | 23 ±2 °C | 2.5 | 2.5 | 3 | 3 | 3 | 3.5 |
| Hardness | Hardness | VF-A01 | N/A | C | 23 ±2 °C | Refer to specific requirement per type of Blown Rubber Outsole | | | | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm3 | 23 ±2 °C | Refer to specific requirement per type of Blown Rubber Outsole | | | | | |
| | DIN abrasion | VF-C01 | N/A | mm$^3$ | 23 ±2 °C | < 300 | < 200 | < 200 | < 150 | < 130 | < 130 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Ground contact | | | | | Sandal construction | < 700 | < 700 | < 500 | 300 | n/a | n/a |
| | Coefficient of Friction – TM144 | VF-B25 | SATRA TM144 | COF | Dry * | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥0.40 | ≥0.40 |
| | | | | | Wet * | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥0.30 |

**Notes:**

A- UV light test is for light colors.

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> If specific gravity varies by more than ± 0.02 from standard, then the material should be rejected

>>> If Durometer hardness varies by more than ± 3 from standard, then the material should be rejected

* For slip resistance specification, a higher standard may be required for specific purposes of use.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.10 Foxing Tape Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels – Minimum Requirement | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm2 | 23 ±2 °C | 90 | 90 | 90 | 90 | 120 | 120 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 325 | 325 | 325 | 325 | 325 | 325 |
| | Die C Tear | VF-B05 | ASTM D624-00(2020) | kgf/cm | 23 ±2 °C | 25 | 25 | 25 | 25 | 30 | 30 |
| Colorfastness | UV- Light [A] | VF-E03 | N/A | AATCC scale | 24hrs, 50 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale [A] | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | rating | 72 hrs., 70 °C | No change on physical appearance | | | | | |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | rating | 7days, 70 °C, 95% RH | No change on physical appearance | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hour, 70 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Bond strength to Upper | VF-F01 | N/A | kgf/cm | 23 ±2 °C | 2.5 | 2.5 | 3 | 3 | 3 | 3.5 |
| | Foxing Tape to Osnaburg | VF-F04 | N/A | kgf/cm | | 2 | 2 | 2 | 2.5 | 2.5 | 2.5 |
| Hardness | Hardness | VF-A01 | N/A | A | 23 ±2 °C | Refer to specific requirement per type of Foxing tape-rubber | | | | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm3 | 23 ±2 °C | Refer to specific requirement per type of Foxing tape-rubber | | | | | |

**Notes:**

A- For light colors.

***For printed foxing tape, refer to "Material Specification for Ink and Printed Material "***

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> If specific gravity is *beyond the standard range of* ± 0.02, then the material should be rejected

>>> If Durometer hardness is *beyond the standard range of* ± 3, then the material should be rejected

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.11 EVA Midsole Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm$^2$ | Material with hardness > 30 C | 12 | 12 | 13 | 15 | 16 | 16 |
| | | | | | Material with hardness 25- 30 C | 10 | 10 | 10 | 12 | 14 | 14 |
| | | | | | Material with hardness < 25 C | 8 | 8 | 8 | 10 | 12 | 12 |
| | Elongation | VF-B01 | N/A | % | All material regardless of hardness | ≥100 | ≥100 | ≥100 | ≥100 | ≥120 | ≥120 |
| | Die C Tear | VF-B05 | N/A | kgf/cm | Material with hardness > 30 C | 5 | 5 | 7 | 8 | 8 | 8 |
| | | | | | Material with hardness 25- 30 C | 4 | 4 | 4 | 6 | 6 | 6 |
| | | | | | Material with hardness < 25 C | 3 | 3 | 3 | 4 | 4 | 4 |
| | Split Tear | VF-B06 | N/A | kgf/cm | Material with hardness > 30 C | 1.5 | 2 | 2 | 2.5 | 3.0 | 3.0 |
| | | | | | Material with hardness 25- 30 C | 1.2 | 1.5 | 1.5 | 1.8 | 2.5 | 2.5 |
| | | | | | Material with hardness < 25 C | 0.75 | 0.75 | 0.75 | 1.20 | 2.0 | 2.0 |
| Compression Set | Static Compression Set | VF-B10 | N/A | % | Material with hardness > 30 C | ≤ 75 | ≤75 | ≤70 | ≤60.0 | ≤50.0 | ≤50.0 |
| | | | | | Material with hardness ≤ 30 C | ≤ 80 | ≤80 | ≤75 | ≤70 | ≤ 60.0 | ≤ 60.0 |
| Resilience | Resilience | VF-B16 | ISO 4662:2017 | % | 23 ±2 °C | 35.0 | 35.0 | 35.0 | 40.0 | 40.0 | 50.0 |
| Flexing | Ross Flex | VF-D02 | N/A | % growth | 23 ±2 °C /100K | 50% | 50% | 50% | 50% | 50% | 50% |
| Colorfastness | Crocking [A] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC scale | Dry | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 |
| | | | | | Wet | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | Perspiration [A] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | UV- Light [B] | VF-E03 | N/A | | 24 hrs., 50 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV- Light [C] | VF-E03 | N/A | | 12 hrs., 50 °C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming / Physical appearance didn't hydrolyze | | | | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | % / rating | 70 °C,95%RH/7 days | 80% retention of tensile strength value before hydrolysis aging / Physical appearance didn't hydrolyze | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 60 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Peel Strength to Rubber | VF-F02 | N/A | kgf/cm | Material with hardness ≥ 30 C | 2.5 | 2.5 | 3 | 3 | 3.5 | 3.5 |
| | | | | | Material with hardness < 30 C | 2 | 2 | 2.5 | 2.5 | 3 | 3 |
| Hardness | Hardness | VF-A01 | N/A | C | 23 ±2 °C | Refer to specific requirement per type of midsole / footbed | | | | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm$^3$ | 23 ±2 °C | Refer to specific requirement per type of midsole / footbed | | | | | |

**Notes:**

**A**- For material with direct skin contact

**B**- UV light test is for light colors.

**C**- For neon color only

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> If specific gravity is ***beyond the standard range of*** ± 0.02, then the material should be rejected

>>> If Durometer hardness is ***beyond the standard range of*** ± 3, then the material should be rejected

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.12 EVA Outsole Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels – Minimum Requirement | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm$^2$ | 23 ±2 °C | 15 | 15 | 15 | 15 | 16 | 17 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | ≥150 | ≥150 | ≥150 | ≥150 | ≥150 | ≥150 |
| | Die C Tear | VF-B05 | N/A | kgf/cm | 23 ±2 °C | 5 | 8 | 8 | 8 | 8 | 8 |
| | Split Tear | VF-B06 | N/A | kgf/cm | 23 ±2 °C | 1.5 | 2 | 2 | 2.5 | 3.0 | 3.0 |
| Compression Set | Static Compression Set | VF-B10 | N/A | % | 6 hrs., 45°C | ≤ 80.0 | ≤70 | ≤70 | ≤60.0 | ≤50.0 | ≤50.0 |
| Resilience | Resilience | VF-B16 | ISO 4662:2017 | % | 23 ±2 °C | 35.0 | 35.0 | 35.0 | 40.0 | 40.0 | 50.0 |
| Flexing | Ross Flex | VF-D02 | N/A | % growth | 23 ±2 °C /100K | 50% | 50% | 50% | 50% | 50% | 50% |
| Colorfastness | UV- Light [A] | VF-E03 | N/A | AATCC scale | 24 hrs., 50 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV- Light [C] | VF-E03 | N/A | | 12 hrs., 50 °C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | rating | 70 °C, 95% RH, 7days | Physical appearance didn't hydrolyze | | | | | |
| | | | | % | | 80% retained of the original strength | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 60 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Peel Strength to Rubber | VF-F02 | N/A | kgf/cm | 23 ±2 °C | 2.5 | 2.5 | 3 | 3 | 3.5 | 3.5 |
| Hardness | Hardness | VF-A01 | N/A | C | 23 ±2 °C | Refer to specific requirement per type of EVA | | | | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm$^3$ | 23 ±2 °C | Refer to specific requirement per type of EVA | | | | | |
| Ground contact | Seam Stitch Tear - Soling | VF-B20 | N/A | kgf/cm | 23 ±2 °C | 27 | 27 | 27 | 27 | 27 | 27 |
| | DIN abrasion [B] | VF-C01 | N/A | mm$^3$ | 23 ±2 °C | <400 | <350 | <350 | <300 | <300 | <250 |
| | Coefficient of Friction – TM144 [B] | VF-B25 | SATRA TM144 | COF | Dry * | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 |
| | | | | | Wet * | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 |

Notes:

A- UV light test is for light colors.

B- For component intended for outsoles.

C- For neon color only

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> If specific gravity is ***beyond the standard range of*** ± 0.02, then the material should be rejected

>>> If Durometer hardness is ***beyond the standard range of*** ± 3, then the material should be rejected

* For slip resistance specification, a higher standard maybe required for specific purposes of use.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.13 TPU Outsole Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels – *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm$^2$ | 23 ±2 °C | 125 | 125 | 125 | 150 | 150 | 200 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 325 | 325 | 325 | 325 | 325 | 325 |
| | Die C Tear | VF-B05 | ASTM D624-00(2020) | kgf/cm | 23 ±2 °C | 50 | 50 | 50 | 75 | 75 | 75 |
| Flexing | Ross Flex | VF-D02 | N/A | % growth | 23 °C /100K | < 50% | < 50% | < 50% | < 50% | < 50% | < 50% |
| | | | | | -15°C / 100K | < 100% | < 100% | < 100% | < 100% | < 100% | < 100% |
| | | | | | -25°C / 100K | -- | -- | -- | -- | -- | < 100% |
| Colorfastness | UV- Light [A] | VF-E03 | N/A | AATCC scale | 24 hrs., 50 C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | rating | 7days, 70 °C, 95% RH | No change on physical appearance | | | | | |
| | | | | % | | 80% retained of the original strength | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hour, 70 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Peel Strength to Upper | VF-F02 | N/A | kgf/cm | 23 ±2 °C | 2.5 | 2.5 | 3 | 3 | 3 | 3.5 |
| Hardness | Hardness | VF-A01 | N/A | A | 23 ±2 °C | Refer to specific requirement per type of TPU Outsole | | | | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm$^3$ | 23 ±2 °C | Refer to specific requirement per type of TPU Outsole | | | | | |
| Ground contact | DIN abrasion | VF-C01 | N/A | mm$^3$ | 23 ±2 °C | < 150 | < 150 | < 120 | < 120 | <100 | <100 |
| | Coefficient of Friction -TM144 | VF-B25 | SATRA TM144 | COF | Dry * | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥ 0.40 | ≥0.40 |
| | | | | | Wet * | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥ 0.30 | ≥0.30 |
| | Non- Marking | VF-E21 | N/A | rating | 23 ±2 °C | non- marking (5.0) | | | | | |

**Notes:**

A- UV light test is for light colors.

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> If specific gravity is ***beyond the standard range of*** ± 0.02, then the material should be rejected

>>> If Durometer hardness is ***beyond the standard range of*** ± 3, then the material should be rejected

* For slip resistance specification, a higher standard maybe required for specific purposes of use.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.14 PU Footbed Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm2 | Footbed with hardness > 30C | 10 | 10 | 12 | 12 | 14 | 14 |
| | | | | | Footbed with hardness 25-30C | 8 | 8 | 10 | 10 | 12 | 12 |
| | | | | | Footbed with hardness ≤ 25C | 6 | 6 | 8 | 8 | 10 | 10 |
| | Elongation | VF-B01 | N/A | % | All materials regardless of hardness | ≥ 250 | ≥ 250 | ≥ 250 | ≥ 250 | ≥ 250 | ≥ 250 |
| | Die C Tear | VF-B05 | N/A | kgf/cm | Footbed with hardness > 30C | 6 | 6 | 8 | 8 | 10 | 10 |
| | | | | | Footbed with hardness 25-30C | 4 | 4 | 6 | 6 | 7 | 7 |
| | | | | | Footbed with hardness ≤ 25C | 3 | 3 | 5 | 5 | 6 | 6 |
| | Split Tear | VF-B06 | N/A | kgf/cm | Footbed with hardness > 30C | 1.2 | 1.2 | 1.5 | 1.5 | 2.0 | 2.0 |
| | | | | | Footbed with hardness 25-30C | 1.0 | 1.0 | 1.2 | 1.2 | 1.5 | 1.5 |
| | | | | | Footbed with hardness ≤ 25 C | 0.75 | 0.75 | 1.0 | 1.0 | 1.2 | 1.2 |
| | Static Compression Set | VF-B10 | N/A | % | 6 hrs., 45 °C | ≤20 | ≤20 | ≤20 | ≤20 | ≤20 | ≤20 |
| | Resilience | VF-B16 | ISO 4662:2017 | % | 23 ±2 °C | 30 | 30 | 30 | 30 | 40 | 40 |
| Flexing | Ross Flex | VF-D02 | N/A | % growth | 23 ±2 °C /100K | 50% | 50% | 50% | 50% | 50% | 50% |
| Colorfastness | Crocking [A] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC scale | Dry | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 |
| | | | | | Wet | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | Perspiration [A] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | UV- Light [B] | VF-E03 | N/A | | 24 hrs., 50 °C | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | | | |
| | Hydrolysis Test / Hydrolytic Stability | VF-E19 | SATRA TM344 | rating | 7days, 70 °C, 95% RH | No deterioration of physical appearance | | | | | |
| | | | | % | | 80% retained of the original strength | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 60 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Peel Strength to Rubber | VF-F02 | N/A | kgf/cm | Footbed with hardness≥ 30 C | 2.5 | 2.5 | 3 | 3 | 3 | 3 |
| | | | | | Footbed with hardness< 30 C | 2 | 2 | 2.5 | 2.5 | n/a | n/a |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm3 | 23 ±2 °C | Refer to specific requirement per type of PU Footbed | | | | | |

**Notes:**

**A-** For material with direct skin contact

**B –** UV light test is for white and light colored only

Specific Gravity (VF- A04) as specified

>>> If specific gravity varies by more than ± 0.02 from standard, then the material should be rejected

>>> If Durometer hardness varies by more than ± 3 from standard, then the material should be rejected

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.15 Sock liner (EVA – Latex – PU) Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | |
| | | | | | | EVA | Latex | PU [E] |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm2 | 23 ±2 °C | 5 | 1.8 | 8 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 75 | 120 | 180 |
| | Die C Tear | VF-B05 | N/A | kgf/cm | 23 ±2 °C | 2 | 1.2 | 2 |
| | Split Tear | VF-B06 | N/A | kgf/cm | 23 ±2 °C | 0.75 | n/a | 0.75 |
| | Static Compression Set | VF-B10 | N/A | % | 6 hrs., 45 °C | ≤ 60 | ≤ 30 | ≤ 20 |
| Resilience | Resilience | VF-B16 | ISO 4662:2017 | % | 23 ±2 °C | 50.0 | n/a | 35.0 |
| Colorfastness | UV- Light [B] | VF-E03 | N/A | AATCC scale | 24 hrs., 50 °C | 3.5 | n/a | 3.5 |
| | Crocking [A] | VF-E02 | AATCC TM8-2016e(2022)e | | Dry | 4.0 | 4.0 | 4.0 |
| | | | | | Wet | 3.5 | 3.5 | 3.5 |
| | Perspiration [A] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4.0 | n/a | 4.0 |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 60 °C | ≤ 2 | ≤ 2 | ≤ 2 |
| Hardness | Hardness | VF-A01 | N/A | C | 23 ±2 °C | Refer to specific requirement per type of sock liner | 30+/-3C | Refer to specific requirement per type of sock liner |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20; ASTM D3574-17 | g/cm3 | 23 ±2 °C | Refer to specific requirement per type of sock liner | 0.28+/-0.02 | Refer to specific requirement per type of sock liner |
| Adhesive Strength | Laminar Peel Strength | VF-F04 | N/A | kgf/cm | MD & DC sock liner: Dry | 0.7 | n/a | 0.7 |
| | | | | | MD & DC sock liner: Wet | 0.5 | n/a | 0.5 |
| | | | | | Latex sock liner: Dry | n/a | 0.35 | n/a |
| | | | | | Latex sock liner: Wet | n/a | 0.25 | n/a |
| | | VF- F05 | N/A | Rating | Dry | No delamination of ink | | |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | EVA | Latex | PU [E] |
| Ink Adhesion on Substrate | Ink Adhesion on Substrate | | | | Wet | No delamination of ink | | |
| | | | | | Artificial Perspiration | No delamination of ink | | |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | Rating | Dry @ 100 cycles | No delamination of ink | | |
| | | | | | Wet @ 50 cycles | No delamination of ink | | |
| | | | | | Artificial Perspiration @ 50 cycles | No delamination of ink | | |
| | Martindale | VF-C04 | N/A | Rating | Dry @ 3,000 cycles | No delamination of ink | | |
| | | | | | Wet @ 1,000 cycles | No delamination of ink | | |
| | | | | | Artificial Perspiration @ 1,000 cycles | No delamination of ink | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | Rating | 70 °C, 95% RH / 7 days | No delamination of ink | | |
| | Heat Aging | VF-E04 | N/A | Rating | 72 hrs., 70° C | No delamination of ink | | |
| | pH Value [C] | | ISO 3071, ISO 4045 | | 23 ±2 °C | Textile:4.0-7.5, natural leather:3.2-7.5 | | |
| | Flammability Test [D] | | CAN/CGSB 4.2 No. 27.5 | | | >4seconds | | |

**Notes:**

**A-** For sock liner with or without top cloth

**B-** UV light test is for light colors. This test is optional for sock liner, unless required by the brand.

C- pH value just for natural leather and textile sock liner cover, pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

**D-** Flammability test MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

E- Ortholite foam please follow individual requirement from GMS Polymer matrix

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified in every type of sock-liner material.

>>> If specific gravity is ***beyond the standard range of*** ± 0.02, then the material should be rejected

>>> If Durometer hardness is ***beyond the standard range of*** ± 3, then the material should be rejected

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.16 PU Foam Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | |
| | | | | | | Low | Medium | High |
| Hardness | Hardness [A] | VF-A01 | N/A | F | 23 ±2 °C | Follow individual requirement from supplier | | |
| Specific Gravity | Specific Gravity [B] | VF-A04 | ASTM D3574 | g/cm3 | 23 ±2 °C | Please refer to footnote [B] | | |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm2 | 23 ±2 °C | 0.9 | 1.2 | 2.2 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 100 | 100 | 100 |
| | Split Tear | VF-B06 | N/A | kgf/cm | 23 ±2 °C | 0.2 | 0.2 | 0.25 |
| | Die C Tear | VF-B05 | N/A | kgf/cm | 23 ±2 °C | 0.3 | 0.6 | 0.9 |
| | Compression Set | VF-B10 | N/A | % | 22 hrs., 23 °C | ≤ 20 | ≤12 | ≤12 |
| Colorfastness | UV- Light [C] | VF-E03 | N/A | AATCC scale | 24 hrs., 50° C | 4.0 | 4.0 | 4.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | rating | 7days, 70 °C, 95% RH | Physical appearance of PU didn't hydrolyze | | |
| | | | | % | | 80% retention of tensile strength value before hydrolysis aging | | |

**Notes:**

**A-** If Durometer hardness varies by more than ± 3 from standard, then the material should be rejected

**B**- Specific gravity: refer to specification by the supplier

**C-** UV light test is for light colors, for exposure part. Any material which has been approved at levels below 3.5 should not be used in exposure parts.

Other material didn't show in manual please refer to specification by the supplier.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.17 Webbing Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | ASTM D6775-13(2024) | kgf | 23 ±2 °C | 25.0 | 30.0 | 35.0 | 35.0 | 45.0 | 45.0 |
| | Stitch Tear | VF-B08 | SATRA TM 33 | kgf/cm | 23 ±2 °C | 8.0 | 10.0 | 10.0 | 10.0 | 12.0 | 12.0 |
| Colorfastness | Water | VF-E01 | N/A | AATCC scale | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Perspiration [A] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [B] | VF-E03 | N/A | | 24 hrs., 50° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV-light [C] | VF-E03 | N/A | | 12 hrs., 50° C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Hydrolysis | VF-E19 | SATRA TM344 | | 7 days / 70 °C / 95% RH | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | rating | | No deterioration of physical appearance | | | | | |
| Water Contact | Wicking | VF-E14 | N/A | mm | Non-wicking/2 hours | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm |
| | pH Value [D] | VF-E21 | ISO 3071 | | 23 ±2 °C | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 |

**Notes:**

**A**- Perspiration test is for direct skin contact only.

**B**- UV light is for light colors.

**C**- For neon color only.

**D**- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.18 Lace (Cotton – Synthetic) Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | SATRA TM94 | kgf | 23 ±2 °C | 25 | 25 | 35 | 40 | 50 | 60 |
| | Tensile After Abrasion | VF-B01 | SATRA TM94 | kgf | 23 ±2 °C | 20 | 20 | 28 | 32 | 40 | 48 |
| | Knot Slip Resistance | VF-B23 | SATRA TM 195 | kgf | 23 ±2 °C | 2.25 | 2.25 | 2.25 | 2.25 | 2.25 | 2.25 |
| | Lace Tip Retention | VF-B24 | SATRA TM 175 | kgf | 23 ±2 °C | 15 | 15 | 15 | 15 | 15 | 15 |
| Abrasion [A] | Lace Abrasion | VF-C09: Method 1 | N/A | cycles | Polyester & Regular (PASS Rating 3) | 2.5K | 5K | 5K | 5K | 15K | 20K |
| | | VF-C09: Method 2 | N/A | cycles | Polyester & Regular (PASS Rating 3) | 2.5K | 5K | 5K | 5K | 15K | 20K |
| | | VF-C09: Method 3 | N/A | cycles | Polyester & Regular (PASS Rating 3) | 2.5K | 5K | 5K | 5K | 15K | 20K |
| | | VF-C09: Method 4 | N/A | cycles | Polyester & Regular (PASS Rating 3) | 2.5K | 5K | 5K | 5K | 15K | 20K |
| | | VF-C09: Method 1 | N/A | cycles | Organic Cotton (PASS Rating 3) | 2.5K | 2.5K | 2.5K | 2.5K | 15K | 20K |
| | | VF-C09: Method 2 | N/A | cycles | Organic Cotton (PASS Rating 3) | 2.5K | 2.5K | 2.5K | 2.5K | 15K | 20K |
| | | VF-C09: Method 3 | N/A | cycles | Organic Cotton (PASS Rating 3) | 2.5K | 2.5K | 2.5K | 2.5K | 15K | 20K |
| | | VF-C09: Method 4 | N/A | cycles | Organic Cotton (PASS Rating 3) | 2.5K | 5K | 5K | 5K | 15K | 20K |
| Colorfastness | Water | VF-E01 | N/A | AATCC scale | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | AATCC scale | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [B] | VF-E03 | N/A | AATCC scale | 24 hrs., 50 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV-light [C] | VF-E03 | N/A | AATCC scale | 12 hrs., 50 °C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| Water Contact | Wicking | VF-E14 | N/A | mm | Non-wicking/2 hours | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 |
| | pH Value [D] | VF-E21 | ISO 3071 | | 23 ±2 °C | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 |

**Notes:**

**A- Lace Abrasion Ratings:**

| Rating | Description |
|---|---|
| 1 | No Visible Wear |
| 2 | Slight dulling or matting of top surface with no pilling, whisking or fraying |
| 3 | Up to 50% wear through top jacket surface with visible pilling, whisking or fraying |
| 4 | > 50% to 100% wear through top jacket surface |
| 5 | Lace broken in two |

B- UV light test is for light colors.

C- For neon color only

D- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.19 Hook & Loop (Velcro ®) Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Tensile | VF-B01 | ASTM D5035-11(2024) | kgf/cm | 23 ±2 °C | 8.0 | 8.0 | 8.0 | 8.0 | 8.0 | 8.0 |
| | Elongation | VF-B01 | ASTM D5035-11(2024) | % | 23 ±2 °C | 20 | 20 | 20 | 20 | 20 | 20 |
| | Stitch Tear | VF-B08 | SATRA TM 33 | kgf/cm | 23 ±2 °C | 4 | 4 | 4 | 4 | 4 | 4 |
| | Peel Strength | VF-B03 | SATRA TM123 | kgf/cm | Initial | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 |
| | | | | | After 10,000 cycles | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |
| | Shear Strength | VF-B03 | SATRA TM123 | kgf/cm2 | Initial | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| | | | | | After 10,000 cycles | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 |
| Colorfastness | Water | VF-E01 | N/A | AATCC scale | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [A] | VF-E03 | N/A | | 24 hrs.,50°C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV-light [B] | VF-E03 | N/A | | 12 hrs.,50°C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | pH Value [C] | VF-E21 | ISO 3071 | | 23 ±2 °C | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 |

**Note:**

A- UV light is for light colors.

B- For neon color only

C- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.20 Thread Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | | | | | | | |
| | | | | | | *Nylon* | | | | | | *Polyester* | | | | | *Cotton* |
| | | | | | | *M8* | *M20* | *M30* | *M40* | *Embroidery* | *Cup sole* | *M20* | *M30* | *M40* | *Embroidery* | *Cup sole* | *M40* |
| Material Strength [C] | Tensile | VF-B01 | N/A | kgf | 23 ±2 °C | 20.0 | 8.5 | 5.5 | 4.0 | 1.0 | 12.0 | 6.0 | 4.5 | 4.0 | 0.8 | 10.0 | 2.0 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 15-35 | | | | | | | | | | | |
| Colorfastness | Water | VF-E01 | N/A | AATCC scale | 38 ±1 °C, 6 hrs. | 3.5 | | | | | | | | | | | |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | | Dry | 4.0 | | | | | | | | | | | |
| | | | | | Wet | 3.5 | | | | | | | | | | | |
| | Perspiration | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C ,6 hrs. | 3.5 | | | | | | | | | | | |
| | UV-light [A] | VF-E03 | N/A | | 24 hrs.,50 °C | 4.0 | | | | | | | | | | | |
| | UV-light [B] | VF-E03 | N/A | | 12 hrs.,50 °C | 3.0 | | | | | | | | | | | |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4.0 | | | | | | | | | | | |
| Water Contact | Wicking | VF-E14 | N/A | mm | Non-wicking/2 hours | ≦ 10mm | | | | | | | | | | | |

**Notes:**

A- UV light test is for light colors.

B- For neon color only

C- Other thread didn't show in manual please refer to specification by the supplier.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.21 Upper Trim Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Rubber | TPU Film | Silicone | P-LEX |
| Material Strength | Tensile | VF-B01 | N/A | kgf/cm2 | 23 ±2 °C | 90 | 130 | 30 | 90 |
| | Elongation | VF-B01 | N/A | % | 23 ±2 °C | 300 | 300 | 180 | 300 |
| | Die C Tear | VF-B05 | ASTM D624-00(2020) | kgf/cm | 23 ±2 °C | 25 | 35 | 10 | 25 |
| | Stitch Tear [F] | VF-B08 | SATRA TM 33 | kgf/cm | 23 ±2 °C | 4 | 5 | 3 | 3 |
| Flexing | Bally [A] | VF-D01 | ASTM D6182-23 | cycles | 23 ±2 °C | 100K / Rank 1 No damage | | | |
| Abrasion | Martindale [B] | VF-C04 | N/A | cycles | 23 ±2 °C / **Dry** | 3K | 3K | 3K | 3K |
| Colorfastness | UV- Light [C] | VF-E03 | N/A | AATCC scale | 24 hrs., 50 °C | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV- Light [G] | VF-E03 | N/A | | 12 hrs., 50 °C | 3.0 | 3.0 | 3.0 | 3.0 |
| | Contact Discoloration [D] | VF-E05 | N/A | | 7 days, 60 °C | 4.0 | 4.0 | 4.0 | 4.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70 °C | 4.0 | 4.0 | 4.0 | 4.0 |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | rating | 7days, 70 °C, 95% RH | 4.0 | | | |
| | | | | % | | 80% retained of the original strength | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 70 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Adhesive Strength | Bond Strength [E] | VF-F02 | N/A | kgf/cm | 23 ±2 °C | 3 | 3 | 3 | 3 |
| | Laminar Peel Strength | VF-F04 | N/A | kgf/cm | | 2.0 | 2.0 | 2.0 | 2.0 |
| Hardness | Hardness | VF-A01 | N/A | °A | 23 ±2 °C | Refer to specific requirement per type of upper trim | | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm3 | 23 ±2 °C | Refer to specific requirement per type of upper trim | | | |

**Notes:**

A- Bally Flex for direct molding and welding application on the upper material or woven label.

B- Martindale abrasion for direct molding on the upper and label material only

C- UV light test for light colors.

D- Contact discoloration for upper trim bonded with light colored component

E- Bond strength follows the material specification level per category or gender

F- TPU Film no need to test the stitch tear for "No-Sew "component.

G– For neon color only

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> If specific gravity is ***beyond the standard range of ±*** 0.02, then the material should be rejected

>>> If Durometer hardness is ***beyond the standard range of ±*** 3, then the material should be rejected

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.22 Zipper Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Chain Crosswise strength | VF-B17 | ASTM D 2061-07(2021) | kgf | 23 ± 2 ° C | 30 | 45 | 45 | 45 | 60 | 60 |
| | Element Pull-Off | VF-B17 | | kgf | 23 ± 2 ° C | 3.5 | 3.5 | 5 | 5 | 10 | 10 |
| | Element slippage, lengthwise | VF-B17 | | kgf | 23 ± 2 ° C | 2.8 | 2.8 | 3.5 | 3.5 | 5.5 | 5.5 |
| | Top Stop Holding Strength | VF-B17 | | kgf | 23 ± 2 ° C | 9 | 9 | 9 | 9 | 12 | 12 |
| | Bottom Stop Holding Strength, Slider | VF-B17 | | kgf | 23 ± 2 ° C | 9 | 9 | 9 | 9 | 15 | 15 |
| | Bottom Stop Holding Strength, Crosswise | VF-B17 | | kgf | 23 ± 2 ° C | 9 | 9 | 9 | 9 | 12 | 12 |
| | Bottom Stop Holding, Stringer Separation | VF-B17 | | kgf | 23 ± 2 ° C | 6 | 6 | 6 | 6 | 8 | 8 |
| | Bridge Top Stop, Stringer Separation | VF-B17 | | kgf | 23 ± 2 ° C | 3.5 | 3.5 | 4.5 | 4.5 | 6 | 6 |
| | Separable Pull off strength | VF-B17 | | kgf | 23 ± 2 ° C | 6 | 6 | 15 | 15 | 20 | 20 |
| | Fixed Retainer Pull off | VF-B17 | | kgf | 23 ± 2 ° C | 6 | 6 | 15 | 15 | 20 | 20 |
| | Separating Unit, Crosswise | VF-B17 | | kgf | 23 ± 2 ° C | 8 | 8 | 8 | 8 | 10 | 10 |
| | Resistance to twist of Pull and slider | VF-B17 | | kgf | 23 ± 2 ° C | 3 | 3 | 3 | 3.5 | 4 | 4 |
| | Resistance to pull -Off slider - pull | VF-B17 | | kgf | 23 ± 2 ° C | 11 | 11 | 15 | 15 | 20 | 20 |
| | Resistance to angular-pull-of slider pull | VF-B17 | | kgf | 23 ± 2 ° C | 3.5 | 3.5 | 4.5 | 4.5 | 6 | 6 |
| | Holding strength of slider lock | VF-B17 | | kgf | 23 ± 2 ° C | 2 | 2 | 4 | 4 | 5 | 5 |
| | Slide Fasteners to Repeated Opening and Closing | VF-B26 | SATRA TM 50 | cycles | Stringer Parallel | 3,000 | 3,000 | 3,000 | 3,000 | 5,000 | 5,000 |
| | | | | | Stringer Apart | 2,000 | 2,000 | 2,000 | 2,000 | 4,000 | 4,000 |
| | Seam Stitch tear | VF-B19 | SATRA TM 180 | kgf/cm | 23 ± 2 ° C | 3 | 5 | 5 | 5 | 6 | 7.5 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 70 °C | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 | ≤ 2 |
| Color fastness | Water | VF-E01 | N/A | AATCC | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | AATCC | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 3.5 | 3.5 |
| | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV- Light [A] | VF-E03 | N/A | AATCC | 24 hrs., 50° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| Water Contact | Wicking | VF-E14 | N/A | mm | Non-wicking/2 hours | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 | ≤ 10 |
| Chemical Resistance | Corrosion Test [B] | VF- E09 | N/A | rating | 23 ± 2° C / 48 hrs. | No Corrosion | | | | | |
| | pH Value [C] | | ISO 3071 | | 23 ± 2° C | 4.0-7.5 | | | | | |

**Notes:**

A- For plastic element only, also for light colors tape

B- For metal element, stoppers, and sliders

C- pH value just for zipper tape, pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.23 Straps & Plugs Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | | | | |
| | | | | | Strap Location | Rubber / PVC / PU | | | Leather ( Natural / Synthetic) | | | Webbing | | |
| | | | | | | Men's | Women's | Children's | Men's | Women's | Children's | Men's | Women's | Children's |
| Material Strength | Tensile Strength | VF-B01 | N/A | kgf | 23 ±2 °C | 25 | | | n/a | | | n/a | | |
| | Strap Pull Strength | VF-F06 | N/A | kgf | Toe post | 12 | 12 | 10 | 25 | 25 | 20 | 25 | 25 | 25 |
| | | | | | Lateral post | 20 | 20 | 15 | 25 | 20 | 20 | 20 | 20 | 20 |
| | | | | | Medial post | 20 | 20 | 15 | 25 | 20 | 20 | 20 | 20 | 20 |
| | | | | | Back strap | 20 | 20 | 15 | 20 | 20 | 20 | 20 | 20 | 20 |
| | | | | | Vamp type of Strap | 30 | 30 | Toddler:15 Children & Jr:30 | 30 | 30 | Toddler:15 Children & Jr:30 | n/a | | |
| Flexing | Ross Flex | VF-D02 | N/A | % growth | 23 ±2 °C /100K | < 100% | | | n/a | | | n/a | | |
| Colorfastness | Water | VF-E01 | N/A | AATCC scale | 38 ±1 °C, 6 hrs. | n/a | | | 3.5 | | | | | |
| | Perspiration [A] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | n/a | | | 3.5 | | | | | |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | | Dry | n/a | | | 4.0 | | | | | |
| | | | | | Wet | n/a | | | 3.5 | | | | | |
| | UV- Light B | VF-E03 | N/A | | 24hrs, 50 °C | 4.0 | | | | | | | | |
| Heat Resistance | Heat Aging | VF-E04 | N/A | | 72hrs ,70 °C | 4.0 | | | | | | | | |
| | Shrinkage | VF- E13 | N/A | % | 1 hr., 70 °C | ≤ 2 | | | | | | | | |
| | Blooming Test | VF-E17 | SATRA TM361 | rating | 60 °C/3 °C | No blooming | | | n/a | | | n/a | | |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | AATCC scale | 7days ,70 °C, 95% RH | 4.0 | | | | | | | | |
| | | | | % | | 80% retained on the original strength | | | | | | | | |
| Hardness | Hardness | VF-A01 | N/A | A | 23 ±2 °C | Refer to specific requirement per type of Polymer strap | | | n/a | | | n/a | | |
| Specific Gravity | Specific Gravity | VF-A04 | ASTM D792-20 | g/cm3 | 23 ±2 °C | Refer to specific requirement per type of Polymer strap | | | n/a | | | n/a | | |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | | | | |
| | | | | | Strap Location | Rubber / PVC / PU | | | Leather ( Natural / Synthetic) | | | Webbing | | |
| | | | | | | Men's | Women's | Children's | Men's | Women's | Children's | Men's | Women's | Children's |

Notes:

A - For direct skin contact

B- UV light test is for light colors.

Specific Gravity (VF- A04) and Durometer (VF – A01) are as specified

>>> Specific gravity varies by more than ± 0.02 from standard, then the material should be rejected

>>> If Durometer varies by more than ± 3 from standard, then the material should be rejected.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.24 Ink & Print Material Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification All Levels / All Categories *Minimum Requirement* |
|---|---|---|---|---|---|---|
| Colorfastness | UV- Light [A] | VF-E03 | N/A | AATCC scale | 24hrs, 50 °C | 4.0 |
| | UV Light [M] | VF-E03 | N/A | | 12 hrs., 50 C | 3.0 |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | | Dry @ 10 cycles | 4.0 |
| | | | | | Wet @ 10 cycles | Light color:3.5, Dark colr:3.0 |
| | Perspiration [B] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | Light color:3.5, Dark colr:3.0 |
| | Water [C] | VF-E01 | N/A | | 38 ±1 °C, 6 hrs. | Light color:3.5, Dark colr:3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | | 72 hrs., 70° C | 4.0 |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | | 70 °C, 95% RH / 7 days | 4.0 |
| Ink Adhesion on Substrate | Ink Adhesion on Substrate / Scotch Tape Resistance | VF- F05 | N/A | Rating | Dry | No delamination of ink |
| | | | | | Wet | No delamination of ink |
| | | | | | Artificial Perspiration [B] | No delamination of ink |
| | Crocking [F] | VF-E02 | AATCC TM8-2016e(2022)e | Rating | Fabric, Leather (Natural & Synthetic)- printed | |
| | | | | | Dry @ 50 cycles | No damage or No abrasion of ink |
| | | | | | Wet @ 20 cycles | No damage or No abrasion of ink |
| | | | | | Artificial Perspiration [B] @ 20 cycles | No damage or No abrasion of ink |
| | | | | | Rubber and other polymer – printed | |
| | | | | | Dry @300 cycles for Vans printed heel logo | No damage or No abrasion of ink |
| | | | | | Dry @ 20 cycles | No damage or No abrasion of ink |
| | Martindale [G, J] | VF-C04 | N/A | Rating | Fabric, Leather (Natural & Synthetic) – printed. This includes size label | |
| | | | | | Dry @ 1000 cycles | No damage or No abrasion of ink [K] |
| | | | | | Wet @ 300 cycles | No damage or No abrasion of ink [K] |
| | | | | | Wet @ 250cycles for Vans printed upper only | No damage or No abrasion of ink [K] |
| | | | | | Artificial Perspiration [B] @ 300 cycles | No damage or No abrasion of ink [K] |
| | | | | | Rubber and other polymer – printed | |
| | | | | | Dry @1000 cycles for Vans printed heel logo | No damage or No abrasion of ink [K] |
| | | | | | Dry @ 50 cycles | No damage or No abrasion of ink [K] |
| | Ross Flex [D, H, I] | VF-D02 | N/A | Rating | 50K / **No slit to be made (PASS RATE = 2)** | No damage or No delamination of ink |
| | Bally Flex [E, L] | VF-D01 | ASTM D6182-23 | Rating | Dry @ 50K cycles for all category except sandal/slipper @20K cycles for sandal/slipper | No damage or No delamination of ink |
| | | | | | Wet @ 20K cycles or all category except sandal/slipper @10K cycles for sandal/slipper | No damage or No delamination of ink |
| | Hydrolysis Test | VF-E19 | SATRA TM344 | Rating | 70 °C, 95% RH / 7 days | No damage or No delamination of ink |
| | Heat Aging | VF-E04 | N/A | Rating | 72 hrs., 70° C | No damage or No delamination of ink |

©2026 VF Corporation. All Rights Reserved

Internal Use

**Notes:**

A- UV light test for light colors.

B- Direct skin contact

C- For ink printed on upper material

D- For ink printed on bottom or thick materials such as plastics, TPU, TPR, PU, Foxing Tape, etc. Ensure NO slit should be made on the test sample

E- For printed components on flexed area. Can also test for other part if required by the brand.

F- Test sample for crocking intended for ink adhesion test should be flat and width at least 16 mm

G- Test sample for Martindale should be flat and diameter of 44± 1 mm (some machine the diameter is only 38 ± 1 mm this is also acceptable), if print on footbed contact with skin directly, please follow sock liner requirement.

H- For ink on the non-flexed area no need to test "Ross Flex "

I – Rating for Ross Flex Test

| Rating | *Description* |
|---|---|
| 1 | No delamination |
| 2 | slight ink delamination or fine cracking of ink ≤ 20% |
| 3 | Ink delamination or ink cracking ≥ 20% but ≤ 50% |
| 4 | Ink delamination or ink cracking ≥ 50 % but not total delamination |
| 5 | Total Ink delamination |

J – For Martindale Test: # of cycles to examine the sample.

| # of cycle to examine | *Regular Category* | | *TBL Tree & TBL PRO* | |
|---|---|---|---|---|
| | Wet / Perspiration | Dry | Wet / Perspiration | Dry |
| 1st check | 50 cycles | 50 cycles | 50 cycles | 100 cycles |
| 2nd check | 100 cycles | 100 cycles | 100 cycles | 500 cycles |
| 3rd check | 150 cycles | 200 cycles | 200 cycles | 1,000 cycles |
| 4th check | 200 cycles | 500 cycles | 500 cycles | 5,000 cycles |
| 5th check | 250 cycles | 750 cycles | 1,000 cycles | 10,000 cycles |
| 6th check | 300 cycles | 1,000 cycles | 5,000 cycles | 20,000 cycles |
| 7th check | | | 10,000 cycles | 30,000 cycles |

K- For Martindale test: the appearance of the ink has been damaged or the material which the ink was printed is already visible due to abrasion of ink.

L- For Bally Flex: # of cycles to examine the sample.

| # of cycle to examine | *Wet* | *Dry* | # of cycle to examine | *Wet* | *Dry* |
|---|---|---|---|---|---|
| 1st check | 5,000 cycles | 5,000 cycles | 4th check | | 30,000 cycles |
| 2nd check | 10,000 cycles | 10,000 cycles | 5th check | | 40,000 cycles |
| 3rd check | 20,000 cycles | 20,000 cycles | 6th check | | 50,000 cycles |

M- For neon color only

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.25 Insole Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | |
|---|---|---|---|---|---|---|---|
| | | | | | | Non-Woven (Stroble & Lasting construction) | Cellulose Board |
| Material Strength | Tensile | VF-B01 | ASTM D5035-11(2024) | kgf/cm | 23 ±2 °C | 8 | 7.5 |
| | % Elongation | VF-B01 | | % | 23 ±2 °C | ≥ 10 | n/a |
| | Trouser Tear [B] | VF-B07 | ASTM D2261-13(2024) | kgf | 23 ±2 °C | 6 | n/a |
| | Seam Stitch Tear | VF-B19 | SATRA TM180 | kgf/cm | 23 ±2 °C | 10 | N/A |
| | Bursting | VF-B02 | ASTM D3786/D3786M-18(2023) | kgf/cm2 | 23 ±2 °C | 10 | n/a |
| | Ross Flex | VF-D02 | N/A | rating | 23 ±2 °C /100K | n/a | ≤ 3 |
| | | | | | -15°C /100K | n/a | ≤ 3 |
| | | | | | -25°C /100K | n/a | ≤ 3 |
| | Laminar Peel Strength | VF-F04 | N/A | kgf/cm | 23 ±2 °C | 0.5 | 0.5 |
| Heat Resistance | Shrinkage | VF- E13 | N/A | % | 1 hr., **60 °C** | 2 | 2 |
| Water Contact | Static Water Absorption | VF-E10 | N/A | % | 23 ±2 °C | <175 | <100 |
| | Water Swelling | VF-E18 | N/A | % | 23 ±2 °C | n/a | ≤ 3 |
| | pH Value [A] | VF-E21 | ISO 3071 | | 23 ±2 °C | 4.0-7.5 | n/a |

**Note:**

A- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

B- For lasting construction non-woven insole no need conducts related test

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.26 Elastic Cord Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels – Minimum Requirement | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic / Classic Boots Outdoor | Adult Performance |
| Material Strength | Tensile | VF-B01 | ASTM D6775-13(2024) | kgf | 23 ±2 °C | 15.0 | 15.0 | 15.0 | 15.0 | 15.0 | 15.0 |
| | Stitch Tear [C] | VF-B08 | SATRA TM 33 | kgf/cm | 23 ±2 °C | 8.0 | 8.0 | 8.0 | 8.0 | 8.0 | 8.0 |
| | Limit of Useful Extension (%) | N/A | SATRA TM102 | % | 23 ±2 °C | 90.0 | 90.0 | 90.0 | 90.0 | 90.0 | 90.0 |
| | Resistance to Repeated Extension | N/A | SATRA TM103 | rating | 23 ±2 °C | Repeated Extension 5,000 cycles | | Repeated Extension 7500 cycles | | | |
| | | | | | | 10% with no more than slight visible damage | | | | | |
| | Resistance to Repeated Extension after heat ageing | N/A | ISO 20870 followed by TM 103 | rating | 72 hrs., 70° C | Repeated Extension 5,000 cycles | | Repeated Extension 7500 cycles | | | |
| | | | | | | 20% with no more than slight visible damage | | | | | |
| Colorfastness | Water | VF-E01 | N/A | AATCC scale | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Perspiration [A] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [B] | VF-E03 | N/A | | 24 hrs., 50° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV-light [D] | VF-E03 | N/A | | 12 hrs., 50° C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Hydrolysis | VF-E19 | SATRA TM344 | AATCC scale | 7 days / 70 °C / 95% RH | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | rating | | No deterioration of physical appearance | | | | | |
| Water Contact | Wicking | VF-E14 | N/A | mm | Non-wicking/2 hours | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm |
| | pH Value [E] | VF-E21 | ISO 3071 | | 23 ±2 °C | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 |

**Notes:**

A- Perspiration test is for direct skin contact only

©2026 VF Corporation. All Rights Reserved



Internal Use

B- UV light is for light colors
C- Only for stitch construction
D- For neon color only
E- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing

## 4.27 Wrapping Paper Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels<br>Minimum Requirement<br>Wrapping Paper |
|---|---|---|---|---|---|---|
| Colorfastness | Crocking [A] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC scale | Dry | 4.0 |
| | PVC Migration [B] | VF-E06 | ISO 15701:2022 | AATCC scale | 50±2°C, 16 hrs | 4.0 |
| | Contact Discoloration [A] | VF-E05 | N/A | AATCC scale | 23 ±2 °C | 4.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs, 70° C | 4.0 |
| | Hydrolysis | VF-E19 | SATRA TM344 | AATCC scale | 7 days / 70 °C / 95% RH | 4.0 |
| | | | | Rating | | No deterioration of physical appearance |
| Ink Adhesion on Substrate | Ink Adhesion on Substrate / Scotch Tape Resistance [A] | VF- F05 | N/A | Rating | Dry | No delamination of ink |
| | | | | | Wet [B] | No delamination of ink |

Note:

A -For printed wrapping paper.
B- For PE wrapping paper only

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.28 Carton – Shoe Box Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | |
|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | |
| | | | | | | Inner box | Outer carton |
| Material Strength | Bursting | VF-B02 | N/A | kgf/cm2 | 23 ±2 °C | 8 | 10 |
| | Edge Crush Test | VF- B27 | SATRA TM440 | kgf/m | 23 ±2 °C | n/a | 500 |
| | Compression | | ASTM D642 | kg | 23 ±2 °C | 80 | n/a |
| Adhesive Strength | Adhesion Strength: Dry Test Method | VF-F11 | N/A | kgf/cm | 23 ±2 °C | n/a | 0.2 kgf/cm |
| | Adhesion Strength: Wet Test Method | VF-F11 | N/A | Rating | 23 ±2 °C | n/a | Lamination shall tear – NOT peel- off |
| Colorfastness | Crocking [A] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC scale | Dry | 4.0 | n/a |
| | Rub Fastness [A] | VF-E24 (GB) | N/A | AATCC scale | Dry (50 cycles @ 1kg load) | 4.0 | n/a |
| | Contact Discoloration [A] | VF-E05 | N/A | AATCC scale | 23 ±2 °C | 4.0 | n/a |

**Notes:**
**A**- For ink coating or printed box

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.29 Woven Label Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Stitch Tear | VF-B08 | SATRA TM 33 | kgf/cm | 23 ±2 °C | 8.0 | 10.0 | 10.0 | 10.0 | 12.0 | 12.0 |
| Colorfastness | Water | VF-E01 | N/A | AATCC scale | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Crocking | VF-E02 | AATCC TM8-2016e(2022)e | | Dry | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | | | | | Wet | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | Perspiration [A] | VF-E16 | AATCC TM15-2021 | | 38 ±1 °C, 6 hrs. | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 | 3.5 |
| | UV-light [B] | VF-E03 | N/A | | 24 hrs., 50° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | UV-light [C] | VF-E03 | N/A | | 12 hrs., 50° C | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 | 3.0 |
| Heat Resistance | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| | Hydrolysis | VF-E19 | SATRA TM344 | | 7 days / 70 °C / 95% RH | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 | 4.0 |
| Water Contact / Water Management | Wicking | VF-E14 | N/A | mm | Non-wicking/2 hours | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm | ≤10 mm |
| | pH Value [D] | VF-E21 | ISO 3071 | | 23 ±2 °C | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 | 4.0-7.5 |

**Notes:**

A – Perspiration test is for direct skin contact only

B – UV light is for light colors.

C – For neon color only.

D – Ph Value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.30 Reinforcement Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Part of Component | Condition | | Material Specification Levels – Minimum Requirement | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Woven-16P,18P,4Oz,10Oz drill | Woven-6Oz,8Oz,10Oz,210D Nylon | Non-Woven | Tricot |
| Material Strength [D] | Tensile | VF-B01 | ASTM D5035-11(2024) | kgf/cm | ALL Parts | All Directions | All Thickness | 3.0 | 5.0 | 6.0 | N/A |
| | Elongation | VF-B01 | ASTM D5035-11(2024 | % | ALL Parts | All Directions | All Thickness | 10 | 10 | 10 | N/A |
| | Trouser Tear | VF-B07 | ASTM D2261-13(2024) | kgf | ALL Parts | All Directions | All Thickness | 1.5 | 3.0 | 4.0 | N/A |
| | Stitch Tear [C] | VF-B08 | SATRA TM 33 | kgf/cm | ALL Parts | All Directions | All Thickness | 2.0 | 3.0 | 5.0 | N/A |
| | Bursting [B] | VF-B02 | ASTM D3786/D3786M-18(2023) | $kgf/cm^2$ | ALL Parts | All Directions | All Thickness | 8 | 10 | 10 | N/A |
| Abrasion | Martindale [A] | VF-C04 | N/A | cycles | ALL Parts | Dry | All Thickness | 12.5K | 25K | 25K | N/A |
| | | | | | | Wet | All Thickness | 3.5K | 6.5K | 6.5K | N/A |
| Color fastness | Crocking [GP] | VF-E02 | AATCC TM8-2016e(2022)e | AATCC Scale | ALL Parts | Dry | colors | 3.5 | 3.5 | 3.5 | 3.5 |
| | | | | | | Wet | colors | 3.0 | 3.0 | 3.0 | 3.0 |
| | Perspiration | VF-E16 | AATCC TM15-2021 | | ALL Parts | 38±1°C,6 hrs. | colors | 3.5 | 3.5 | 3.5 | 3.5 |
| Water Contact | Wicking | VF-E14 | N/A | mm | ALL Parts | Non-wicking/2hrs | All Thickness | ≤10 | ≤10 | ≤10 | ≤10 |
| Fabric Density | Yarn Count | VF-A07 | ASTM D3775-17(2023) | # of yarns | ALL Parts | Length | All Thickness | For woven fabric, refer to material specification per material | | | |
| | | | | | ALL Parts | Width | All Thickness | | | | |
| | Weight | VF-A02 | ASTM D3776/D3776M-20(2025) | $g/m^2$ | ALL Parts | 23 ±2 °C | All Thickness | Refer to material specification per material | | | |
| | pH Value [E] | | ISO 3071 | | ALL Parts | 23 ±2 °C | All Thickness | 4.0-7.5 | | | |
| | Flammability test [F] | | CAN/CGSB 4.2 No. 27.5 | | ALL Parts | 23 ±2 °C | All Thickness | >4seconds | | | |

©2026 VF Corporation. All Rights Reserved

Internal Use

**Notes:**

A- Martindale abrasion only for skin contact material.

B- Material bursting is acceptable below specs if it stretches 1.0 cm or more above the top clamping ring without breaking or bursting

C- Reinforcement material without stitching in finish shoes no need to test the stitch tear

D- For single layer reinforcement material with lower material strength, the physical properties will be checked again with laminated material.

E- pH value testing MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

F- Flammability test MUST be arranged in 3rd party lab as it is considered as a safety requirement in some locations. The material is supposed to be tested together with the RSL testing.

**Other reinforcement material didn't show in manual please refer to specification by the supplier.**

GP **– Guidelines on Proper use:**

>>> Crocking and colorfastness – any material which have been approved at levels below 3.5 Dry and 3.0 Wet should not be used in contact with light colored materials or materials with a significant contrast in color.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.31 Shoe Charms Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Strength | Pull Strength | VF-B04 | SATRA TM149 | kgf | 23±2°C | 9.5 | | | | | |
| Heat Resistance | Hydrolysis | SATRA TM344 | SATRA TM344 | rating | 7days, **70°C** 95 % RH | No Loss of Stones or inlays or Damage / 4.0G | | | | | |
| | Heat Aging | VF-E04 | N/A | rating | 72hrs, 70 °C | No Loss of Stones or inlays or Damage / 4.0G | | | | | |
| | Washing Test | VF-G03 | N/A | | | No Loss of Stones or inlays or Damage | | | | | |
| Colorfastness | UV- Light [A] | VF-E03 | N/A | rating | 24hrs, 50 °C | 4.0G | | | | | |
| | UV- Light [B] | VF-E03 | N/A | rating | 12hrs, 50 °C | 3.0G | | | | | |
| Chemical Resistance | Corrosion Test | VF-E09 | N/A | rating | 23±2°C / 48 hrs. | No Corrosion | | | | | |
| | Security of attachment of stones or inlays | | SATRA TM275 | rating | | No Loss of Stones or inlays or Damage | | | | | |
| Safety Requirement | Sharp Points [C] | | CFR 1500.48 | rating | | No Sharp Points | | | | | |
| | Sharp Edges [C] | | CFR 1500.49 | rating | | No Sharp Edges | | | | | |
| | Small Parts [C] | | CFR 1501 | rating | | No Small Parts | | | | | |

**Note:**

A - UV light test is for white and light colors.
B - For neon colors only
C - For Kid's products, arranged in 3rd party lab.

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.32 Bonding Strength Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Description to be Bonded | Component | Material Specification Levels *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength [B] | Bond Strength | VF-F01 | N/A | kgf/cm | Perimeter / Area Pull [A] | Upper to Midsole | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Midsole to Midsole | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 | 2.5 |
| | | | | | | Midsole to Outsole | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Upper to Outsole | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Sandal construction - foot bed to midsole [C, D] | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| | | | | | | Upper to Foxing tape | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Foxing tape to Outsole | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Logo to foxing tape/outsole/midsole | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Foxing Tape to Foxing Tape [E] | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Toe bumper to Foxing tape [E] | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | Rubber Toe Cap to Upper | 1.5 | 1.5 | 2.0 | 2.0 | 2.0 | 2.0 |
| | | | | | | Insole to filler | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| | | | | | | Filler to outsole | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| | | | | | | TPU / TPR/ Nylon / Gel to Outsole | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | | TPU / TPR/ Nylon / Gel to Midsole | 2.5 | 2.5 | 3.0 | 3.0 | 3.5 | 3.5 |
| | | | | | Perimeter bond after washing / aging / hydrolysis [F] | All components | 80% retained of the original strength | | | | | |
| | Laminar Peel Strength | VF-F04 | N/A | kgf/cm | Perimeter / Area / Whole unit | Upper to reinforcement/ laminated Toe box/RB/PU/TPU | 1.5 | 1.5 | 1.5 | 2.0 | 2.0 | 2.0 |
| | | | | | | Upper to molded / injection logo / non-sew logo / HF logo | 1.5 | 1.5 | 1.5 | 2.0 | 2.0 | 2.0 |

©2026 VF Corporation. All Rights Reserved



Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Description to be Bonded | Component | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| | | | | | | Upper / Lining to non-sew component | 1.5 | 1.5 | 1.5 | 2.0 | 2.0 | 2.0 |
| | | | | | | Lining to non- sew logo | 0.5 | 0.5 | 0.5 | 0.5 | 0.7 | 0.7 |
| | | | | | | Foxing Tape to Osnaburg | 2.0 | 2.0 | 2.0 | 2.5 | 2.5 | 2.5 |

**Notes:**

**A-** Soft midsole, EVA, Gel will require OVERRIDE specs. Brand developers should issue and sign the required bond strength

**B-** If result of bonding strength is below standard, but the failure in the material – result must meet the material specs requirement

**C**- For sandal construction only, foot-bed to midsole/outsole if some style without midsole /next layer(arch/wedge)

**D**- For sandal construction, with hardness below 25 °C, should follow the footnote specified in **C**

**E, F**- Optional, only if needed

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.33 Bonding Strength Tolerance

| Test | Description of Test | Component | Material Specification Levels | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | *Minimum Requirement* | | | | | | | | |
| | | | Toddler / Children & Jr | | | Adult Price Value / Adult Casual | | | Athletic / Adult Performance | | |
| | | | Standard bond strength (kgf/cm) | TOLERANCE | | Standard bond strength (kgf/cm) | TOLERANCE | | Standard bond strength (kgf/cm) | TOLERANCE | |
| | | | | 90 % of standard: ALL Points | 80 % of standard: 3 points ONLY | | 90 % of standard: ALL Points | 80 % of standard: 3 points ONLY | | 90 % of standard: ALL Points | 80 % of standard: 3 points ONLY |
| Bond strength [A] | Perimeter / Area Pull | Upper to Midsole | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Midsole to Midsole | 2.5 | 2.25 | 2.0 | 2.5 | 2.25 | 2.0 | 2.5 | 2.25 | 2.0 |
| | | Midsole to Outsole | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Upper to Outsole | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Sandal construction - footbed to midsole [B, C] | 1.5 | no tolerance | no tolerance | 1.5 | no tolerance | no tolerance | 1.5 | no tolerance | no tolerance |
| | | Upper to Foxing tape | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Foxing tape to Outsole | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Logo to foxing tape / outsole / midsole | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Foxing Tape to Foxing Tape [D] | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Toe bumper to Foxing tape [D] | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Rubber Toe Cap to Upper | 1.5 | no tolerance | no tolerance | 2.0 | no tolerance | no tolerance | 2.0 | no tolerance | no tolerance |
| | | Upper to Rand | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |
| | | Insole to Filler | 1.5 | no tolerance | no tolerance | 1.5 | no tolerance | no tolerance | 1.5 | no tolerance | no tolerance |
| | | TPU / TPR/ Nylon / Gel to Outsole | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |

©2026 VF Corporation. All Rights Reserved

Internal Use

| Test | Description of Test | Component | Material Specification Levels | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | *Minimum Requirement* | | | | | | | | |
| | | | Toddler / Children & Jr | | | Adult Price Value / Adult Casual | | | Athletic / Adult Performance | | |
| | | | Standard bond strength (kgf/cm) | TOLERANCE | | Standard bond strength (kgf/cm) | TOLERANCE | | Standard bond strength (kgf/cm) | TOLERANCE | |
| | | | | 90 % of standard: ALL Points | 80 % of standard: 3 points ONLY | | 90 % of standard: ALL Points | 80 % of standard: 3 points ONLY | | 90 % of standard: ALL Points | 80 % of standard: 3 points ONLY |
| | | TPU / TPR/ Nylon / Gel to Midsole | 2.5 | 2.25 | 2.0 | 3 | 2.7 | 2.4 | 3.5 | 3.15 | 2.8 |

Notes:

A- If result of bonding strength is below standard, but the failure in the material – result must meet the material specs requirement

B- For sandal construction only, foot-bed to midsole

C – For sandal construction, with hardness below 25 °C, should follow the footnote specified in **A**

D - Optional, only if needed.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.34 Pull Strength Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Minimum Requirement | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Material Strength | Strength of Hardware Attachment (Eyelet) [A] | VF-B04 | SATRA TM 149 | kgf | 23 ±2 °C | 15.0 | 15.0 | 21.0 | 21.0 | 27.0 | 27.0 |
| | Punched hold | | SATRA TM 149 | kgf | 23 ±2 °C | 15.0 | 15.0 | 21.0 | 21.0 | 27.0 | 27.0 |
| | Strength of Hardware Attachment (Hook, D-Ring, Buckles) | | SATRA TM 149 | kgf | 23 ±2 °C | 21.0 | 21.0 | 35.0 | 35.0 | 45.0 | 45.0 |
| | Strength of Hardware Attachment (Rivet with Knob) | | SATRA TM 149 | kgf | 23 ±2 °C | 9.5 | 9.5 | 9.5 | 9.5 | 18.0 | 18.0 |
| | Cord lock as Lace holder | | SATRA TM 149 | kgf | 23 ±2 °C | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 | 2.0 |
| | Cord lock as lace locker | | SATRA TM 149 | kgf | 23 ±2 °C | 12 | 12 | 12 | 12 | 12 | 12 |
| | Lace end puller | | SATRA TM 149 | kgf | 23 ±2 °C | 15 | 15 | 15 | 15 | 15 | 15 |
| | Snap button | | SATRA TM 149 | kgf | 23 ±2 °C | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 |
| | Strength of Lace Heel Pull Tab [B] | | SATRA TM 149 | kgf | 23 ±2 °C | 25.0 | 30.0 | 35.0 | 35.0 | 45.0 | 45.0 |
| | Strength of Lace Ghillie / Eyerow [BC] | | SATRA TM 149 | kgf | 23 ±2 °C | 25.0 | 30.0 | 35.0 | 35.0 | 45.0 | 45.0 |
| | Strength of Lace Loop Holder [BC] | | SATRA TM 149 | kgf | 23 ±2 °C | 12.0 | 12.0 | 12.0 | 15.0 | 15.0 | 15.0 |
| | Strength of First Lace Loop / Tongue Pull Tab [BC] | | SATRA TM 149 | kgf | 23 ±2 °C | 12.0 | 12.0 | 12.0 | 15.0 | 15.0 | 15.0 |
| | Ornaments attached to upper, strap, rubber, and other components [D] | | SATRA TM 149 | kgf | 23 ±2 °C | 9.5 | 9.5 | 9.5 | 9.5 | 9.5 | 9.5 |

**Notes:**

A - non-metal eyelet should also use this material specification.
B - Other types of lace holder besides metal should use this standard.
C - Webbing or other type lace holder or pull-tab component which is non- functional must also use this standard.
D - non-metal ornaments should also use this material specification.

©2026 VF Corporation. All Rights Reserved

Internal Use

## 4.35 Whole Shoe Performance Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels – *Minimum Requirement* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |
| Footwear Performance | Whole Shoe Flex @ RT | VF-G02 | N/A | cycles | 23 ±2 °C/Flexing angles: Option 1: angle @ 45 degrees for vulcanize casual shoes Option 2: angle @ 30 degrees for thick sole | n/a | 125K/sandals 50K | 125K/sandals 50K | 125K/sandals 50K | 125K | 125K |
| | Whole Shoe Flex @ freezing | | | cycles | -20 °C [A] | n/a | n/a | n/a | n/a | n/a | 100K |
| | Whole Shoe Flex - **Waterproof** [C] | VF-G04 | N/A | cycles | With WP cement/ water level 20mm **above top line of the midsole** | 15K | 15K | 15K | 15K | 15K | 15K |
| | Whole Shoe Waterproof **-Static** | VF-G07 | N/A | hour | 23 ±2 °C | 4 | 4 | 4 | 4 | 4 | 4 |
| | Whole Shoe Wash Test [D] | VF-G03 | N/A | rating | 23 ±2 °C | After 3 cycles: no signs of bleeding / deterioration of properties / delamination, color change is 3G minimum | | | | | |
| | Hydrolysis Resistance | VF-E19 | SATRA TM344 | AATCC scale | 7 days / **70°C** / 95% RH | 4 | 4 | 4 | 4 | 4 | 4 |
| | | | | rating | | No signs of bleeding / deterioration of properties / delamination, metal hardware/component corrosion no evaluation, Nubuck/Suede color change no evaluation | | | | | |
| | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70°C | 4 | 4 | 4 | 4 | 4 | 4 |
| | | | | rating | | No signs of bleeding / deterioration of properties / delamination, hardware corrosion | | | | | |
| | UV- Light [B] | VF-E03 | N/A | AATCC scale | 24 hrs., 50° C | 4 | 4 | 4 | 4 | 4 | 4 |
| | Slip Resistance [E] | VF-B25 | SATRA TM144 | | 23 ±2 °C | Dry:0.4 Wet:0.3 [F] | | | | | |

**Notes:**

A - Freezing temperature may vary as required by the brand.

B - Neon color test time 12hrs, requirement is 3.0G minimum.

C - Gore Tex styles follow Gore Tex requirement directly.

©2026 VF Corporation. All Rights Reserved



Internal Use

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Levels | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Minimum Requirement* | | | | | |
| | | | | | | Toddler | Children & Jr | Adult Price Value | Adult Casual | Adult Athletic | Adult Performance |

D - If the shoe is made of leather + fabric with color block design and the color of leather material falls into the scope below, colorfastness to water on material stage will be reviewed. When test result of color staining is lower than grade 2, whole shoe wash test will be needed. Color'O number 00-40. Pantone code starts with 17 -19.

E- New compound or new tooling with traction function should be tested in development stage, production stage is optional test.

F- This requirement is minimum requirement, if individual compound with a higher standard maybe required, please follow material specification from Polymer matrix.

©2026 VF Corporation. All Rights Reserved



Internal Use

## 4.36 Snow Boots (Vans) Performance Test Requirement

| Specification Property | Test | VF FW Test Method | International Test Method | Units | Condition | Material Specification Level<br>*Minimum Requirement*<br>Snow Boots | |
|---|---|---|---|---|---|---|---|
| Footwear Performance | Whole Shoe Flex @ RT | VF-G02 | N/A | cycles | 23 ±2 °C / Flex angle @ 30 ° / **for flexible bottoms only** | 125K | |
| | Whole Shoe Flex @ freezing | | | cycles | -20 °C | 100K | |
| | Whole Shoe Water Absorption | VF-G06 | N/A | % | 1 hour / 23 ±2 °C / water level: **20 mm above top line of the midsole** | **No water leakage** | |
| Material Strength | Bond strength | VF-F01 | N/A | kgf/cm | Upper to outsole | 4 | |
| | | | | | Upper to non sew / HF component | 2.0 | |
| | | | | | Outsole to Midsole | 3.5 | |
| | Bootie Liner Tongue Pull Strength | VF-B04 | SATRA TM 149 | kgf | 23 ±2 °C | 18 | |
| | Heel Pull Tab | | | | | 100 | |
| | Webbing Harness | | | | | 40 | |
| | Upper Eye -Row | | | | | Toddler | Children & Jr /Adult |
| | Strength of Hardware Attachment (Eyelet) | VF-B04 | SATRA TM 149 | kgf | 23 ±2 °C | 50 | 60 |
| | Strength of Hardware Attachment (Hook, D-Ring, Buckles) | | | | | 50 | 80 |
| | Webbing | | | | | 100 | |
| | Boa Cable Guide [A] | | | | | 100 | |
| | Lower Eye -Row | | | | | | |
| | Strength of Hardware Attachment (Eyelet) | VF-B04 | SATRA TM 149 | kgf | 23 ±2 °C | 60 | |
| | Webbing | | | | | 80 | |
| | Boa Cable Guide [A] | | | | | 80 | |
| Heat Aging Test | Hydrolysis Resistance | VF-E19 | SATRA TM344 | rating | 7 days / 70°C / 95% RH | No peeling off hardware coating and no deformation of material | |
| | | | | AATCC scale | | 4 | |
| | Heat Aging | VF-E04 | N/A | AATCC scale | 72 hrs., 70° C | 4 | |
| | UV- Light | VF-E03 | N/A | | 24 hrs., 50° C | 4 | |
| Chemical Resistance | Corrosion Test | VF- E09 | N/A | rating | 23 ±2 °C/ 48 hrs. | No Corrosion | |

**Note:**

**A-** BOA Cable Guide: Please follow BOA's requirement for different type BOA component

©2026 VF Corporation. All Rights Reserved



Internal Use

## Leather Tannage Specification

TBL/TNF/Vans leather which test requirement set up by GMS team, any new leather is no tannage specification please contact with leather team.

## One VF Polymer Matrix

TBL/TNF/Vans polymer (Rubber/EVA/PU/Insole) test requirement set up by GMS team, any polymer is no requirement please contact with polymer team.

©2026 VF Corporation. All Rights Reserved

Internal Use

# SECTION 5: Sampling for Testing

## 5.1 Mass Production Testing Frequency (Factory In-house lab)

| TEST | DAILY: 1 pair per line | Bi-weekly/Monthly by style / SKU[4] | Quantity |
|---|---|---|---|
| Bonding | Yes | -- | 1 pair |
| Pull test | Yes | -- | 1 pair |
| Waterproof (Flex & Static) | Yes | -- | 1 pair |
| Footwear flexing @ 23° C | -- | Bi-weekly (By Style) | 1 pair |
| Heat aging | -- | Monthly (By SKU) | ½ pair |
| Hydrolysis | -- | Bi-weekly (By SKU) | ½ pair |
| Washing [1] | -- | Monthly (By SKU) | ½ pair |
| UV Light [2] | -- | Bi-weekly (By SKU) | n/a |
| Dip Test [3] | -- | Monthly (By SKU) | 1 pair |

**Notes:**

1. ***Washing Test: For shoes intended to be washed only.***

2. ***UV Light Test: dissect sample from bonding specimen or whole shoe flexing. Extract the sample between upper and midsole or foxing tape to check yellowing from primer and cement Also check the resistance of white & light-colored components against UV light. Size of specimen- refer to Test Method: VF - E03.***

3. ***GB Flex Testing: for China market only.***

**4. If PO quantity less than 500 pairs, it could share test results with other bigger PO which are same style produce in same month.**

| MATERIAL / COMPONENTS | TEST | DAILY | EVERY DELIVERY | QUANTITY |
|---|---|---|---|---|
| Upper & Lining | All physical properties | -- | Yes | 1 complete test per batch/delivery |
| Trims, hardware, closures, other components | All physical properties | -- | Yes | 1 complete test per batch/delivery |
| Reinforcement material | All physical properties | -- | -- | 1 complete test per material /color per month |
| Bottoms: sourced | All physical properties | -- | Yes | 1 complete test per batch/delivery |
| Bottoms: In- house | All physical properties | Yes | -- | 1 complete test per compound/color per week |
| Foxing Tape | All physical properties | Yes | | 1 complete test per compound/color per week |

©2026 VF Corporation. All Rights Reserved


Internal Use

# SECTION 6: Nominated Testing Laboratories

## SGS

| China - Guangzhou | China- Hong Kong | Vietnam- Ho Chi Minh |
|---|---|---|
| **SGS-CSTC Standards Technical Services Co., Ltd.**<br>198 Kezhu Road, Science Park,<br>Guangzhou Economic & Technology<br>Development Zone, Guangzhou,<br>Guangdong, CHINA, 510663<br><br>**CONTACT:** Cassie Dai<br>**T:** +86 (0)20-32136643<br>**F:** +86 (0)20-82075169<br>**E:** cassie.dai@sgs.com | **SGS Hong Kong Ltd.**<br>4/F, On Wui Centre, 25 Lok Yip Road, Fanling, New Territories, Hong Kong<br><br>**CONTACT:** Doris Chou<br>**T:** (852) 23344481 extn:1268<br>**T:** (852)27747133 (sample pick up)<br>**E:** doris.choy@sgs.com | **SGS Vietnam., Ltd.**<br>Lot 3/21, 19/5A Street, Tan Binh IZ, Tay Thanh Ward,<br>Tan Phu District, HCMC, Vietnam<br><br>**CONTACT:** Nga Bui<br>**T:** +8428 38160999 Ext.655<br>**F:** +8428 38160996<br>**E:** nga.bui@sgs.com |
| **Turkey***<br>**SGS Turkey – Istanbul**<br>SGS Supervise Gozetme Etud Kontrol Servisleri A.S.<br>Is Istanbul Plaza Baglar Mah. Osmanpasa Cad. No:95 E Girisi, Gunesli, 34209, Istanbul, Turkey<br>**CONTACT:** Eda Akgül<br>**T:** +90-212-368-4000 (Ext:4570)<br>**F:** +90-212-296-4782<br>**E :**eda.akgul@sgs.com | | |

*Just for Whole shoe test only

## Intertek Testing Services (ITS)

| China - Guangzhou | China- Hong Kong | Mexico-Leon |
|---|---|---|
| **Intertek Testing Services Guangzhou Ltd.**<br>3rd Floor, Hengyun Building,<br>235 Kaifa Avenue,<br>Guangzhou Economic & Technological Development District,<br>Guangzhou 510730, China<br><br>**CONTACT:** Vivian Nie<br>**T:** +86 20 82139277<br>**E:** vivian.nie@intertek.com | **Intertek Testing Services Hong Kong Ltd.**<br>1st Floor, Garment Centre,<br>576 Castle Peak Road,<br>Kowloon, Hong Kong, China<br><br>**CONTACT:** Mr. Keith Lee<br>**T:** +852 98639126<br>**E**: keith.lee@intertek.com | **Intertek Leon, Mexico**<br>Pirul del Valle, No. 113, Col. Valle de León, C.P. 37140, León Guanajuato.<br><br>**CONTACT:** Cecilia de Leon Escobedo<br>**T:** +52 477 774 6938 Ext. 25090072<br>**Tel. Office** +52 477 774 6938<br>**E:** cecilia.deleon@intertek.com |

©2026 VF Corporation. All Rights Reserved


Internal Use

## Intertek Testing Services (ITS)

| | | |
|---|---|---|
| **Turkey*** <br> **Intertek Test Hizmetleri Anonim Şirketi Turkey** <br> Merkez Mahallesi, Sanayi Cad. No.23, Altindag Plaza, Yenibosna-34197, Istanbul, Turkey <br> **CONTACT:** Masuk Turan/ Nihal Akgul <br> **T:** +90 212 496 4699 / 4707 <br> **F:** +90 212 452 80 55 <br> **E:** masuk.turan@intertek.com <br> **E:** nihal.akgul@intertek.com | **Vietnam** <br> **Intertek Testing Services Vietnam Ltd.** <br> 8th Floor, Lobby D, S.O.H.O Biz Office Building, No. 38 Huynh Lan Khanh St., Ward 2, Tan Binh District, Ho Chi Minh City, Vietnam <br> **CONTACT: Ms. Hongnhung Nguyen** <br> **T:** +84 28 6297 1099 – ext. 674 <br> **T:** +84 28 6297 1092 <br> **F:** +84 28 6297 1098 <br> **E:** hongnhung.nguyen@intertek.com | |

*Just for Whole shoe test only

©2026 VF Corporation. All Rights Reserved


Internal Use

# SECTION 7: Change Log

| Version | Date | Revision | Justifications |
|---|---|---|---|
| 01 | Jan 2024 | First version | |
| 02 | Jan 2025 | Second version | 1. Add new requirement for reinforcement material.<br>2. Update requirement for fabric lining.<br>3. Update requirement for ink/print material.<br>4. Update Whole shoe performance test requirement.<br>5. Remove Solvent reaction test requirement.<br>6. Update Straps & Plugs test requirement.<br>7. Add PU footbed test requirement.<br>8. Separate Pull strength test requirement |
| 03 | Jan 2026 | Third version | 1. Change Slip resistance test method to SATRA TM144<br>2. Update Slip resistance test requirement<br>3. Change Hydrolysis test method to SATRA TM344<br>4. Change Blooming test method to SATRA TM361<br>5. Add note for whole shoe washing test<br>6. Add new test requirement for shoe charms<br>7. Add midsole to midsole bonding test requirement<br>8. Update snow boots test requirement |

©2026 VF Corporation. All Rights Reserved


Internal Use